

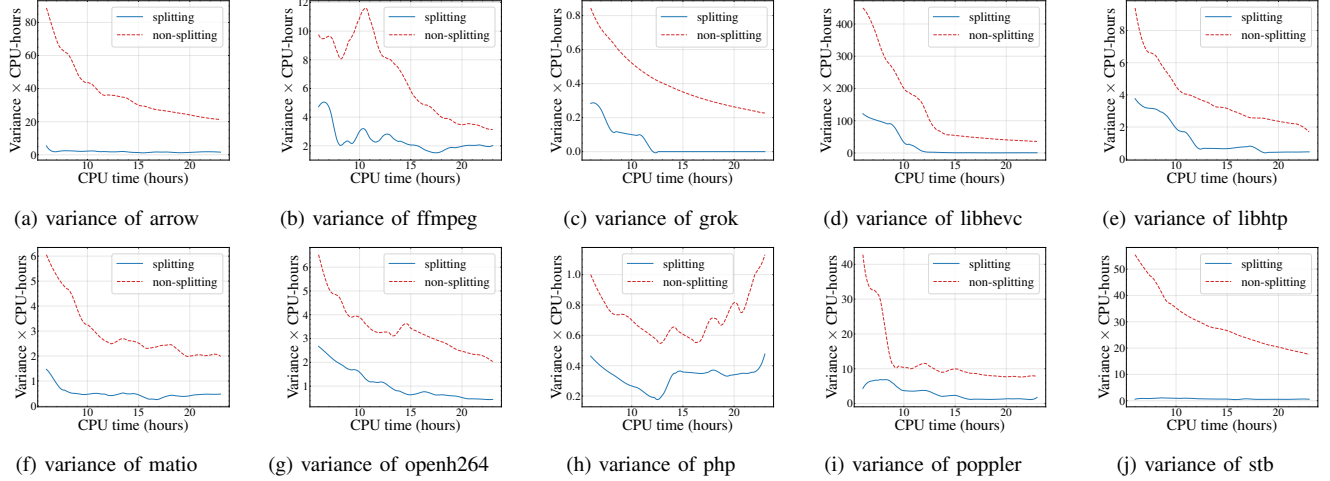


Figure 9: Comparisons of variance across 10 benchmarks for fuzzer Mopt

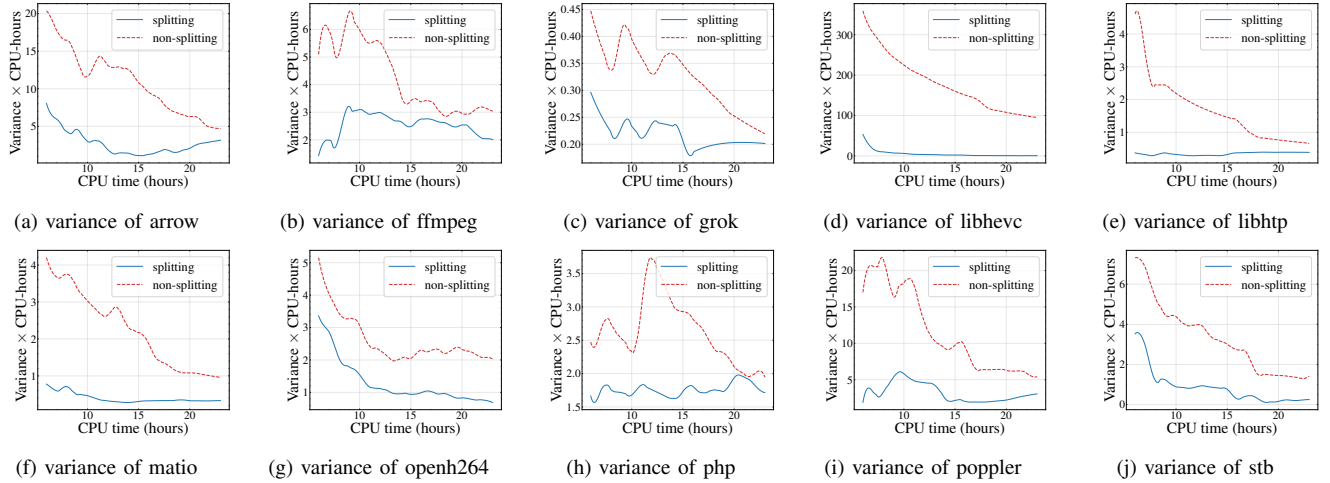


Figure 10: Comparisons of variance across 10 benchmarks for fuzzer Afl

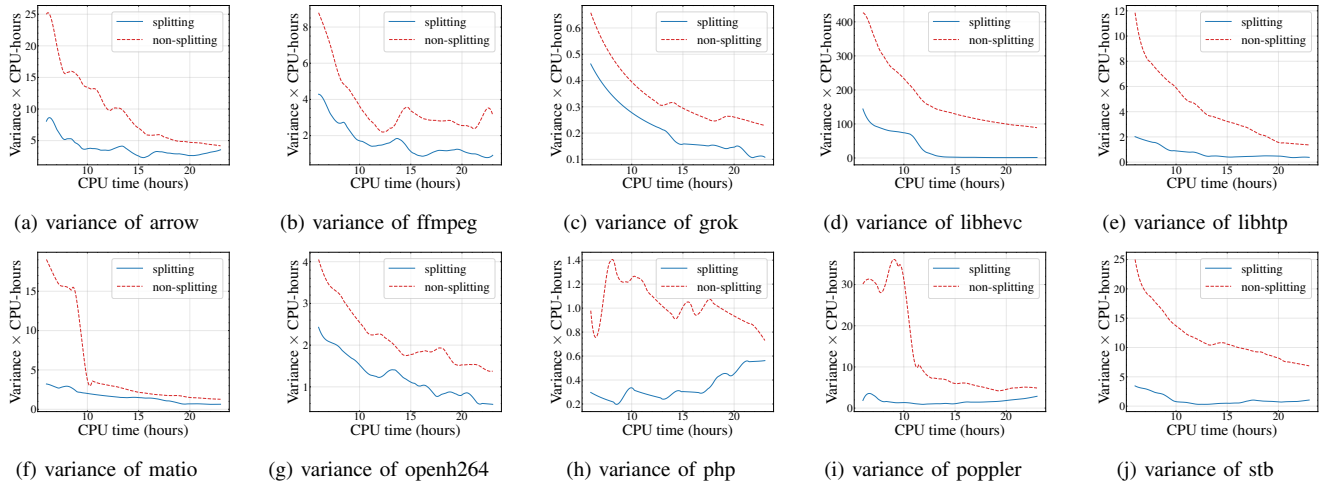


Figure 11: Comparisons of variance across 10 benchmarks for fuzzer Aflsmart

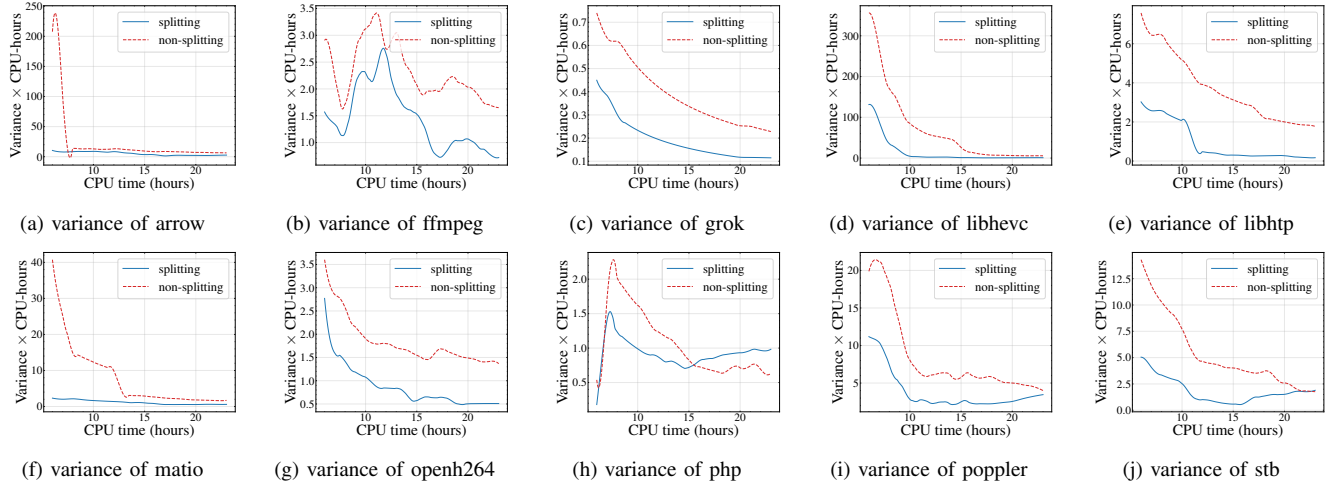


Figure 12: Comparisons of variance across 10 benchmarks for fuzzer Afffast

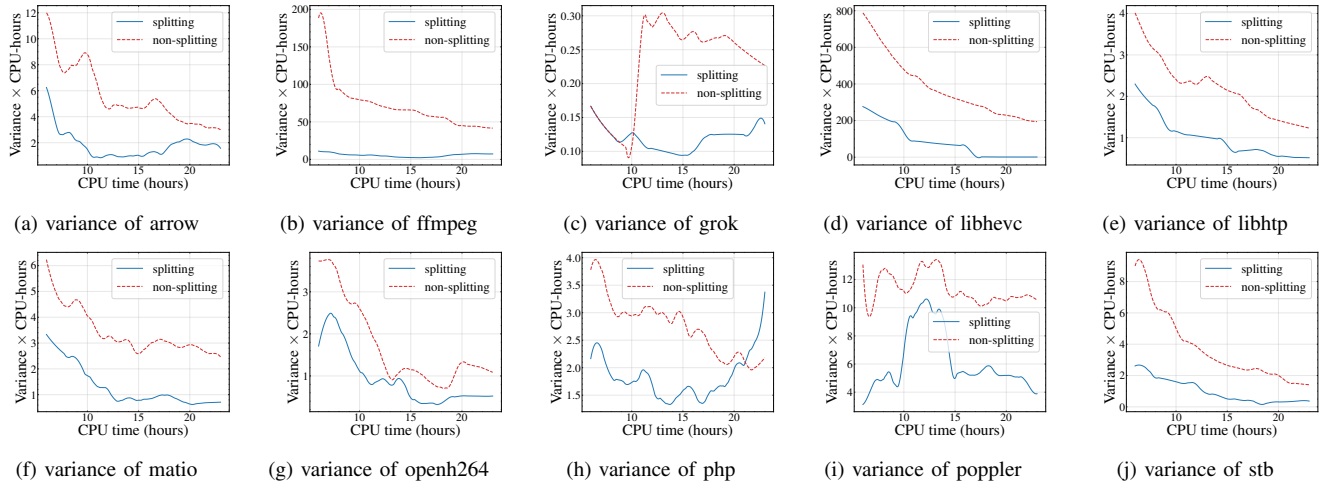


Figure 13: Comparisons of variance across 10 benchmarks for fuzzer Honggfuzz

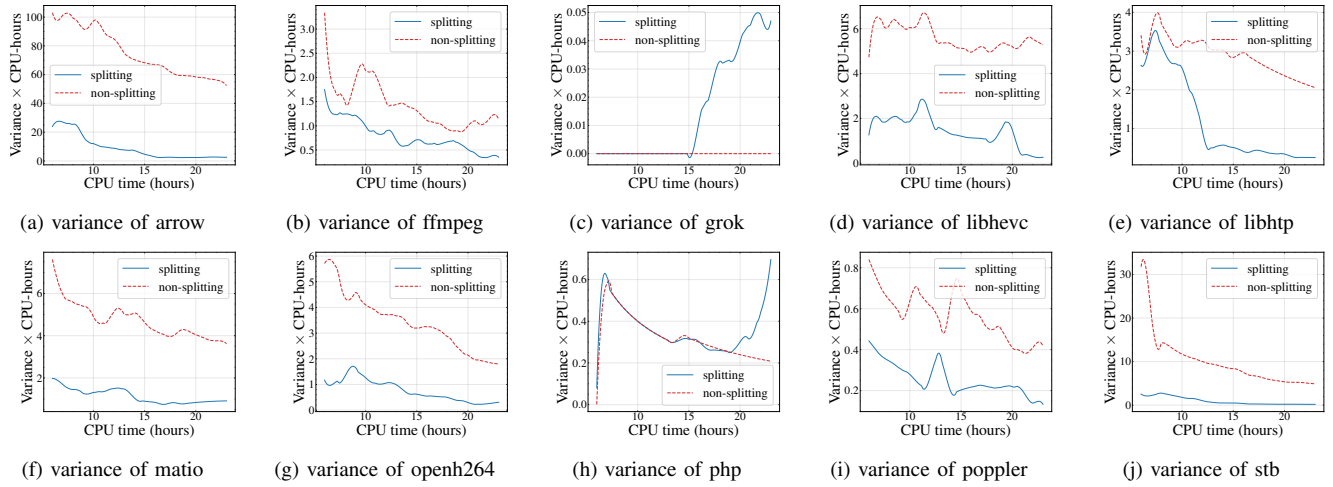


Figure 14: Comparisons of variance across 10 benchmarks for fuzzer Libfuzzer

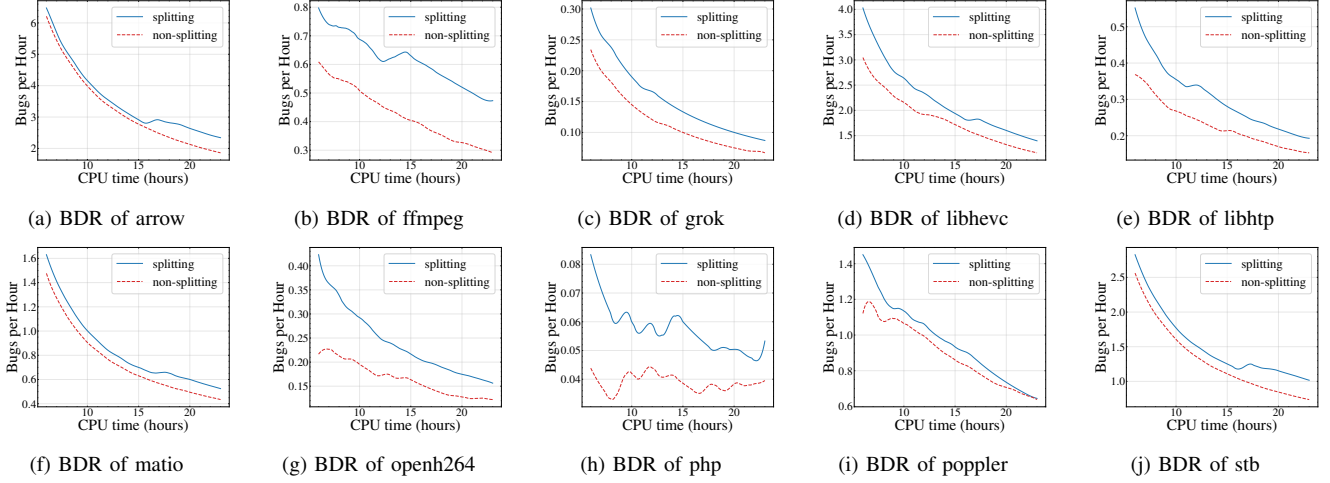


Figure 15: Comparisons of bug detection rate across 10 benchmarks for fuzzer Mopt

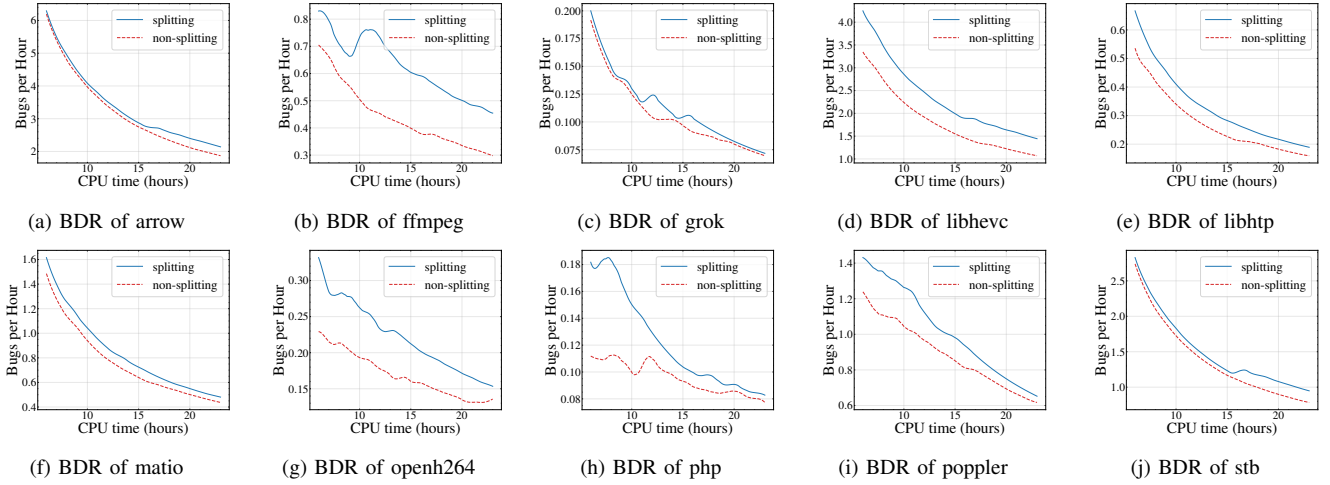


Figure 16: Comparisons of bug detection rate across 10 benchmarks for fuzzer Afl

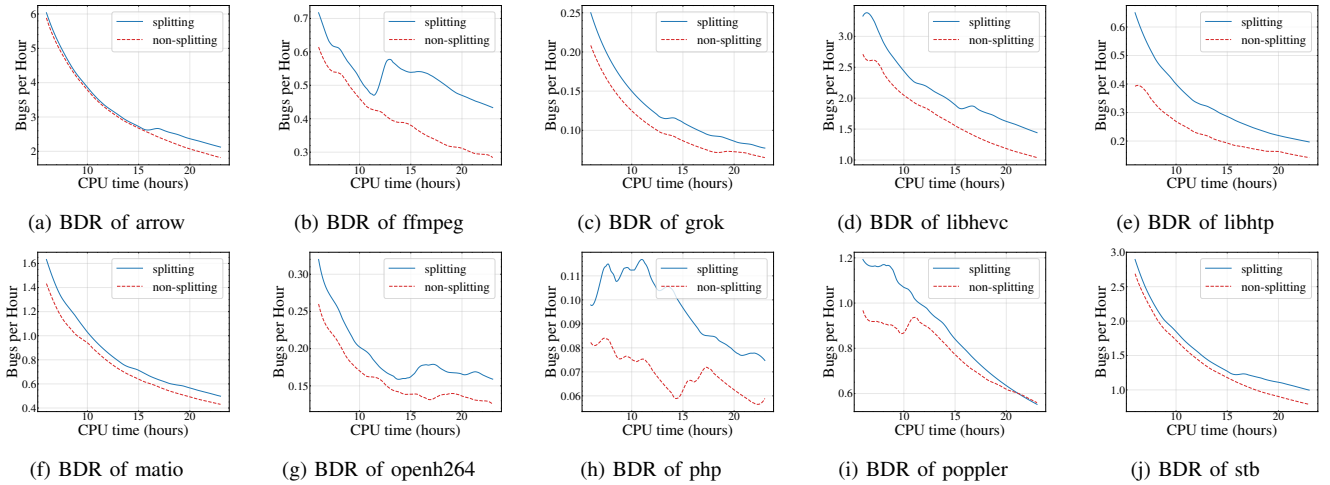


Figure 17: Comparisons of bug detection rate across 10 benchmarks for fuzzer Aflsmart

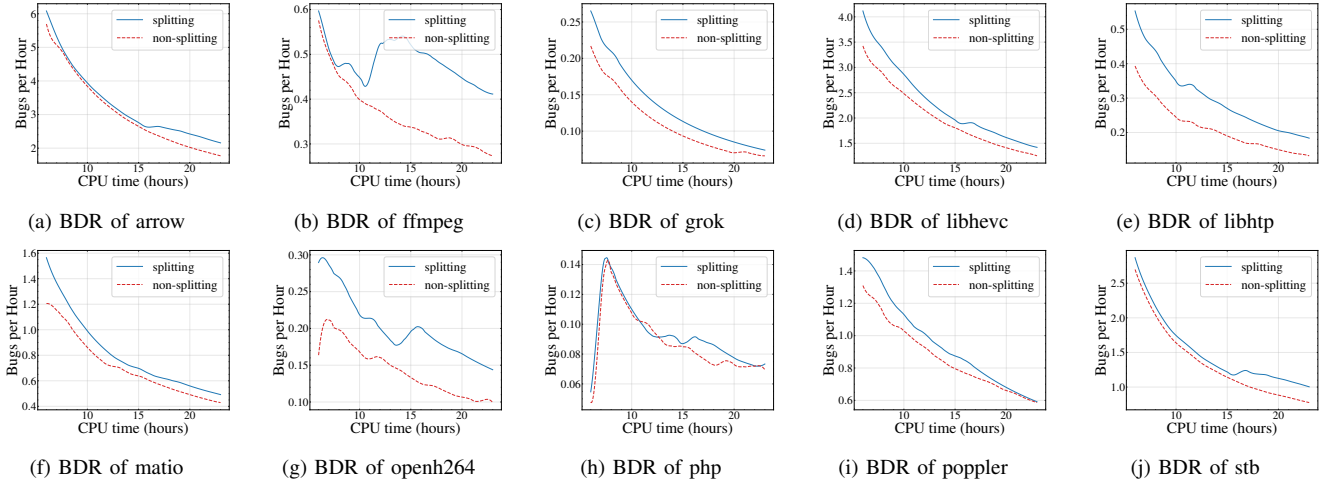


Figure 18: Comparisons of bug detection rate across 10 benchmarks for fuzzer Aflfast

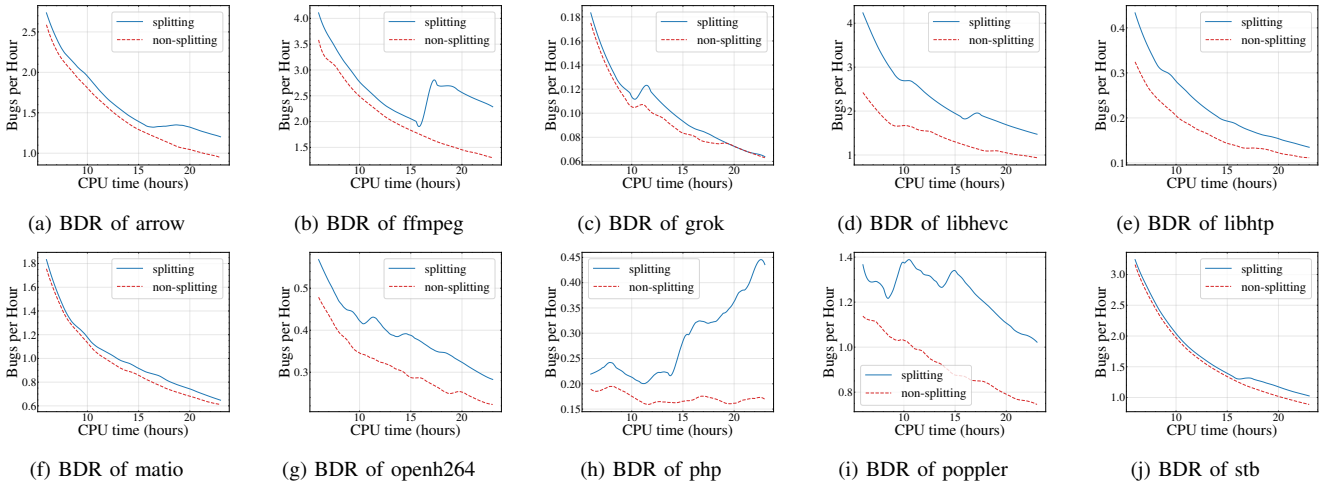


Figure 19: Comparisons of bug detection rate across 10 benchmarks for fuzzer Honggfuzz

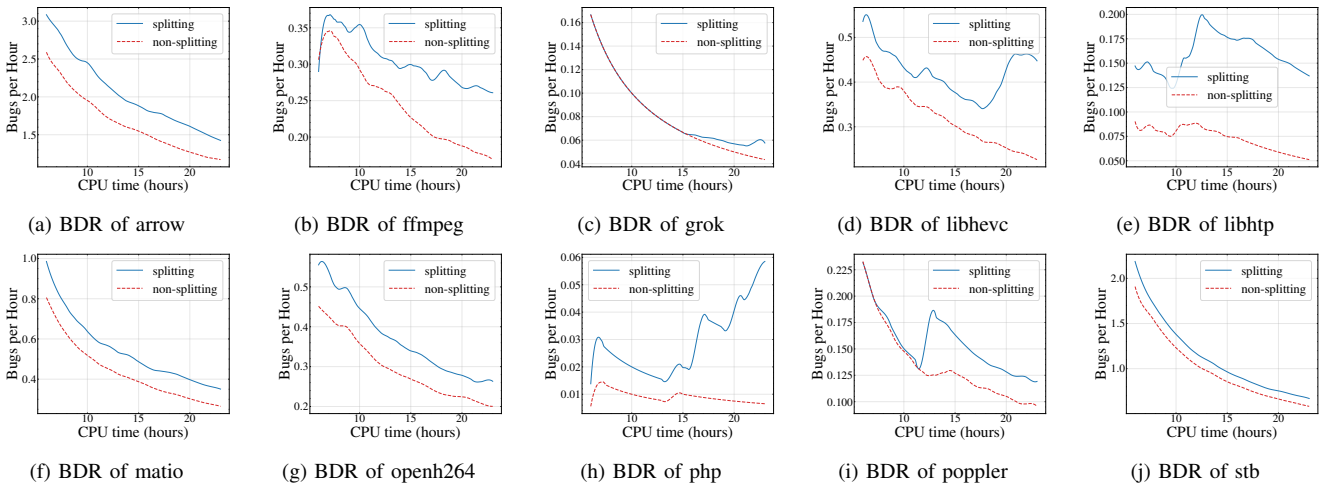


Figure 20: Comparisons of bug detection rate across 10 benchmarks for fuzzer Libfuzzer

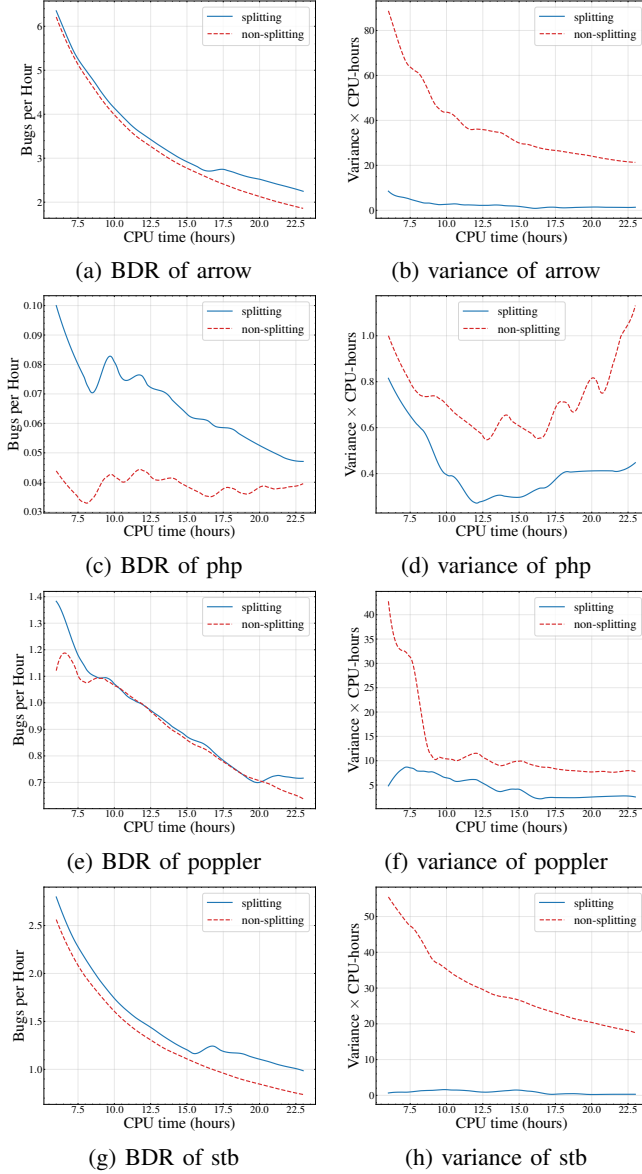


Figure 21: Ablation study for fuzzer Mopt on arrow, php, poppler and stb

For the MC estimator, we first consider the case where $M = 1$. Based on the aforementioned derivations, we obtain that

$$\begin{aligned}
 & \text{Var}(\sqrt{T}\tilde{P}_R(T)) \\
 &= \sigma^2 - 2 \sum_{t=1}^{\infty} \gamma_t + 2 \sum_{t=1}^{(T-1)} (1 - \frac{t}{T}) \gamma_t \\
 &= \sigma^2 - 2 \sum_{t=1}^{T-1} \gamma_t - 2 \sum_{t=T}^{\infty} \gamma_t + 2 \sum_{t=1}^{(T-1)} (1 - \frac{t}{T}) \gamma_t \\
 &= \sigma^2 - 2 \sum_{t=1}^{(T-1)} (\frac{t}{T}) \gamma_t - 2 \sum_{t=T}^{\infty} \gamma_t.
 \end{aligned}$$

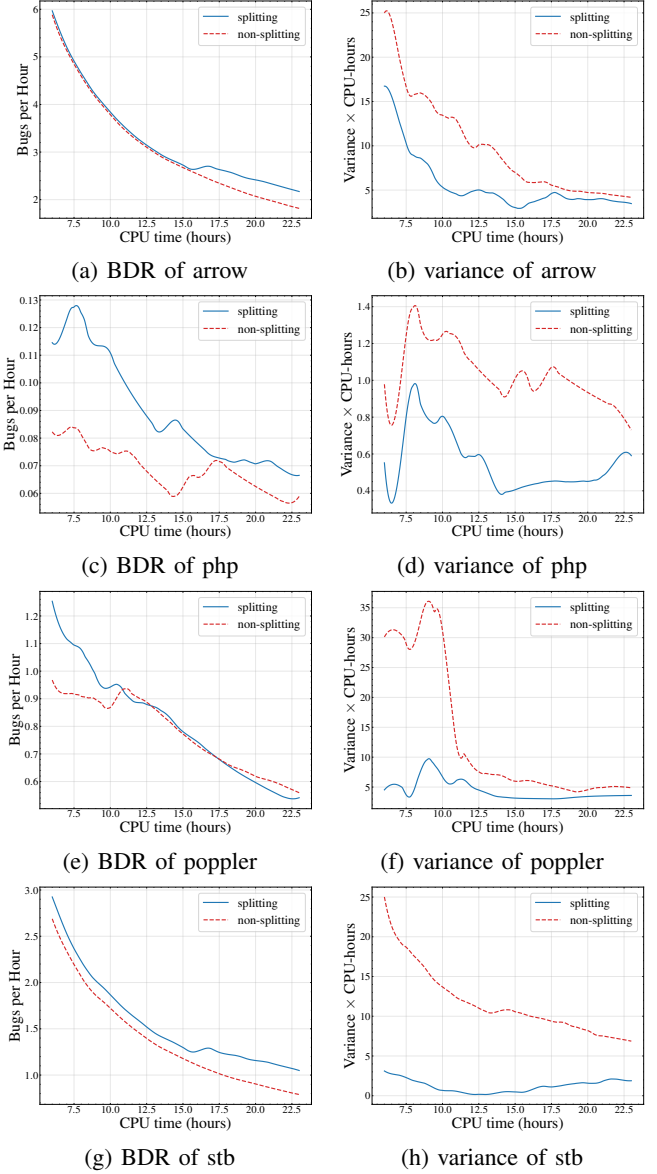


Figure 22: Ablation study for fuzzer Aflsmart on arrow, php, poppler and stb

Moreover, for a geometrically ergodic Markov chain, we have that

$$\gamma_t = \frac{\pi(1)(1 - \gamma_{M,R})^t}{2 \max \{\pi(0), \pi(1)\}}$$

which immediately implies that

$$\begin{aligned}
 & \frac{2}{T} \sum_{t=1}^{\infty} t \cdot \gamma_t \\
 &= \frac{2}{T} \sum_{t=1}^{\infty} t \cdot \frac{\pi(1)(1 - \gamma_{M,R})^t}{2 \max \{\pi(0), \pi(1)\}} \\
 &= \frac{2}{T} \frac{\pi(1)}{2 \max \{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2}
 \end{aligned}$$

and

$$\begin{aligned} & \sum_{t=T}^{\infty} \gamma_t \\ &= \sum_{t=T}^{\infty} \frac{\pi(1)(1 - \gamma_{M,R})^t}{2 \max\{\pi(0), \pi(1)\}} \\ &= \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1}{\gamma_{M,R}} \end{aligned}$$

Subsequently, we arrive at

$$\begin{aligned} & \text{Var}(\sqrt{T} \tilde{P}_R(T)) \\ &= \sigma^2 - \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} - \\ & \quad 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1}{\gamma_{M,R}} \end{aligned}$$

and let us denote

$$\sigma_{MC}^2 = \text{Var}(\tilde{P}_R(T)).$$

For the splitting algorithm with a single split, the estimator variance satisfies

$$\text{Var}(\hat{P}_R(T)) = \text{Var}\left(\frac{1}{T} \sum_{t=1}^T \frac{1}{k_t^m} \sum_{s=1}^{k_t^m} Y_{s,t}\right).$$

As before, we first focus on the case $M = 1$, i.e., a single initial trial. Suppose the run splits at time τ and then evolves with two active paths thereafter; specifically, $k_t = 1$ for $t \leq \tau$ and $k_t = 2$ for $t > \tau$. This corresponds to a single binary split (as opposed to multiple splits). In the multi-replication setting, the split time τ need not be synchronized across replications.

5.1. Assumptions. To characterize the variance of the *splitting estimator*, we adopt the following assumptions. We model the fuzzing process as a (possibly non-stationary) Markov chain on a state space with a designated bug (event) set B (denoted a in the proof). In particular, after a split, we treat the descendant paths as independent paths initiated from the bug-triggering state, while allowing dependence within each descendant path.

A1 (Fuzzer dynamics: AR (1) model). The state evolution within each post-split replication is modeled by an autoregressive (order-one) update of the form

$$U_{k,t} = V(U_{k,t-1}) + W_{k,t}, \quad k \in \{1, 2\},$$

where $V(\cdot)$ represents the (possibly randomized) masking/selection operation applied to the previous state, and $W_{k,t}$ represents the injected mutation/input at time t . Specifically,

$$X_{1,t} = \phi(X_{1,t-1}, W_{1,t}, Z_{1,t}) = \phi(X_{1,t-1}) \cdot W_{1,t} + Z_{1,t} \cdot W_{1,t},$$

with $\phi = (\phi_1, \dots, \phi_l)$ denoting masking operators over locations and $W_{k,t}$ denoting location selections.

A2 (Domain-set structures). We treat the bug set B as a discrete set of inputs/states. In addition, we assume that

any bit in the input string can potentially trigger a bug, and that each bug is associated with a specific bit/value pattern.

A3 (Correlation control via a multiplicative bound). We assume that

$$\mathbb{P}(X_{2,t} \in B, X_{1,t} \in B) \leq (1-\rho) \mathbb{P}(X_{1,t} \in B) \mathbb{P}(X_{2,t} \in B).$$

which can be met by the construction of the AR (1) model by the following lemma.

Lemma 5 (Sufficient condition for Assumption A3 under a coupled-noise AR(1) construction). *Consider the post-split inputs $\{X_{k,t}\}_{t \geq \tau+1}$, $k \in \{1, 2\}$ evolving under the AR(1)-type update in Assumption A1, where the mutation at time t selects a location J_t with distribution $\mathbb{P}(J_t = j) = P(j)$ and then injects a value in the domain D_j according to a pmf $f_j(\cdot)$ (allowing non-uniform mutation). Let the bug set B be discrete (Assumption A2), and suppose that membership in B admits the following bitwise description: for each location j there is a (finite) set $V_j \subseteq D_j$ such that “being buggy due to location j ” corresponds to $X_t[j] \in V_j$.*

Assume the coupled-noise construction satisfies the following exclusion property: conditional on $X_{1,t}$ and on selecting $J_t = j$, the second replica draws

$$X_{2,t}[j] \sim f_j(\cdot) \text{ restricted to } D_j \setminus \{X_{1,t}[j]\},$$

i.e., $X_{2,t}[j] \neq X_{1,t}[j]$ almost surely at the mutated location (all other coordinates evolve as in A1).

Then there exists $\rho \in (0, 1]$ such that for every t ,

$$\mathbb{P}(X_{2,t} \in B, X_{1,t} \in B) \leq (1-\rho) \mathbb{P}(X_{1,t} \in B) \mathbb{P}(X_{2,t} \in B),$$

i.e., Assumption A3 holds. Moreover, one may take

$$\rho = 1 - \frac{\sum_j P(j) \sum_{\ell \in V_j \setminus \{x[j]\}} f_j(\ell)}{\sum_j P(j) \sum_{\ell \in V_j} f_j(\ell)},$$

where $x[j]$ denotes the (bug-triggering) value at location j realized by $X_{1,t}$ on the event $\{X_{1,t} \in B\}$. In the special case $|V_j| = 1$ for each j (at most one buggy value per location), one can take $\rho = 1$.

Proof. By the definition of conditional probability,

$$\mathbb{P}(X_{2,t} \in B, X_{1,t} \in B) = \mathbb{P}(X_{1,t} \in B) \mathbb{P}(X_{2,t} \in B \mid X_{1,t} \in B).$$

Fix t . Under the bitwise description, the (marginal) probability that $X_{2,t}$ is buggy can be written as

$$\mathbb{P}(X_{2,t} \in B) = \sum_j P(j) \sum_{\ell \in V_j} f_j(\ell).$$

On the event $\{X_{1,t} \in B\}$, at the bug-relevant location j the coupled-noise exclusion removes the realized value $X_{1,t}[j]$ from the support for $X_{2,t}[j]$. Hence, conditional on $J_t = j$,

$$\mathbb{P}(X_{2,t}[j] \in V_j \mid X_{1,t} \in B, J_t = j) \leq \sum_{\ell \in V_j \setminus \{X_{1,t}[j]\}} f_j(\ell),$$

which is strictly smaller than $\sum_{\ell \in V_j} f_j(\ell)$ whenever $f_j(X_{1,t}[j]) > 0$ and $X_{1,t}[j] \in V_j$. Averaging over j yields

$$\begin{aligned} \mathbb{P}(X_{2,t} \in B \mid X_{1,t} \in B) &\leq \sum_j P(j) \sum_{\ell \in V_j \setminus \{X_{1,t}[j]\}} f_j(\ell) \\ &= (1 - \rho) \mathbb{P}(X_{2,t} \in B) \end{aligned}$$

for ρ as defined in the statement. Multiplying by $\mathbb{P}(X_{1,t} \in B)$ gives

$$\mathbb{P}(X_{2,t} \in B, X_{1,t} \in B) \leq (1 - \rho) \mathbb{P}(X_{1,t} \in B) \mathbb{P}(X_{2,t} \in B),$$

which is exactly Assumption A3. For the special case $|V_j| = 1$, the exclusion implies $\mathbb{P}(X_{2,t} \in B \mid X_{1,t} \in B) = 0$, so $\rho = 1$ works. $\square$

A4 (Geometric ergodicity and mixing approximation)..

Assume that the Markov chain model is geometrically ergodic (under the coupled-noise construction). Consequently, after a burn-in period ℓ that exceeds the mixing time, the marginal probabilities admit approximations of the following form (up to exponentially small errors):

$$\mathbb{P}(U_{1,t} \in B) \approx \mathbb{P}(U_{2,t} \in B) \approx \pi(1).$$

A5 (Rare-bug regime).. Assume that $\pi(1) \ll \pi(0)$, so bugs occur with low probability, consistent with practical settings.

5.2. Stationary Markov Chains with One Split. In this section, we formally compare the *splitting estimator* with the vanilla *MC estimator* for stationary Markov chains under a single split.

Variance comparison.. The variance of the splitting estimator satisfies

$$\begin{aligned} \sigma_{\text{splitting}}^2 &= V_R(T) \\ &= \frac{1}{2T^2} \text{Var}\left(\sum_{t=\tau+1}^T Y_{1,t}\right) + \frac{1}{T^2} \text{Var}\left(\sum_{t=1}^{\tau-1} Y_{s,t}\right) \\ &\quad - \frac{\rho}{2T^2} \cdot \sum_{t=\tau+1}^T \mathbb{P}(X_{1,t} \in B) \mathbb{P}(X_{2,t} \in B). \end{aligned}$$

Moreover, the difference $\sigma_{\text{splitting}}^2 - \sigma_{MC}^2$ is approximately of the order

$$G_T = \frac{\sigma_f^2}{2T} - \frac{\rho}{2T} = \frac{1}{2T}(\sigma_f^2 - \rho),$$

where

$$\sigma_f^2 = \lim_{T \rightarrow \infty} \text{Var}\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T \mathbf{1}_{\{X_t \in B\}}\right),$$

and ρ is defined through the correlation control condition

$$\mathbb{P}(X_{2,t} \in B, X_{1,t} \in B) \leq (1 - \rho) \cdot \mathbb{P}(X_{1,t} \in B) \cdot \mathbb{P}(X_{2,t} \in B).$$

Correlation model and the origin of ρ .. We model the fuzzing process as an autoregressive system of the form

$$\begin{aligned} X_{1,t} &= \phi(X_{1,t-1}, W_{1,t}, Z_{1,t}) = \phi(X_{1,t-1}) \cdot W_{1,t} + Z_{1,t} \cdot W_{1,t} \\ &= \sum_{j=1}^l \phi_j(X_{1,t-1}) W_{1,t}^j + \sum_{j=1}^l Z_{1,t} W_{1,t}^l. \end{aligned} \quad (1)$$

This formulation implies the correlation bound

$$\mathbb{P}(X_{2,t} \in B, X_{1,t} \in B) \leq (1 - \rho) \cdot \mathbb{P}(X_{1,t} \in B) \cdot \mathbb{P}(X_{2,t} \in B).$$

Here, *masking* is represented by $\phi = (\phi_1, \dots, \phi_l)$, where each ϕ_j sets location j to NULL. The *selections* are given by $W_{k,t} = (W_{k,t}^1, \dots, W_{k,t}^l)$ for $k \in \{1, 2\}$, where $W_{k,t}^j \sim \text{Bernoulli}$ indicates selection of location j . Finally, the *inputs* $Z_{1,t}$ and $Z_{2,t}$ are identically distributed (with CDF F and PDF f) but dependent (via $U_{1,t}$ and $1 - U_{1,t}$).

Closed-form expressions.. Consequently, we obtain the following expressions for σ_f^2 and ρ :

$$\sigma_f^2 = \pi(1)\pi(0) + \frac{\pi(1)}{\pi(0)} \cdot \frac{1 - \gamma_{M,R}}{\gamma_{M,R}}, \quad \rho = \frac{1}{|D| |\pi(1)|},$$

where ρ is independent of the transition kernel P_R . In particular, the dominating terms satisfy

$$\sigma_f^2 \approx \frac{\pi(1)}{\gamma_{M,R}}, \quad \rho \approx \frac{1}{|D| |\pi(1)|}.$$

When does splitting increase or reduce variance?. We compare σ_f^2 and ρ through the threshold condition

$$\pi(1) \gtrless \sqrt{|D| \gamma_{M,R}}.$$

If $\pi(1) > \sqrt{|D| \gamma_{M,R}}$ (i.e., $\sigma_f^2 \geq \rho$), then splitting yields a slight variance increase. If $\pi(1) \leq \sqrt{|D| \gamma_{M,R}}$ (small underlying bug rate), then splitting yields smaller variance, demonstrating its advantage in variance reduction. Thus, splitting is particularly beneficial when the program's bug rate $\pi(1)$ is small relative to $\sqrt{|D| \gamma_{M,R}}$.

Sample complexity advantage.. Finally, we compare sample complexities. Without splitting, the sample complexity is $2T$, whereas with splitting it is $2T - \tau$. Since $2T - \tau < 2T$, splitting has lower sample complexity than non-splitting. Overall, splitting can reduce variance in the low-bug-rate regime and also improves sample complexity, supporting further development and adoption of the approach.

5.3. Non-stationary Markov Chains with One Split. We next extend the variance comparison between the splitting-based estimator and the vanilla Monte Carlo (MC) estimator to the setting of non-stationary Markov chains under a single split.

Assumption in the non-stationary regime: controlled deviation and regularity.. For the non-stationary extension, the proof assumes the deviation between the stationary distribution π and the distribution induced by the (time-varying) kernel over a burn-in horizon can be controlled by a total-variation-type bound q . The proof further assumes a regularity condition on the mapping that defines the correlation constant (e.g., $b(\cdot)$ is Lipschitz, motivated by smoothness), allowing the distributional deviation to translate into a deviation bound on ρ of the form $|\rho - \rho_{\text{ref}}| \leq L\delta$, where L depends on the transition kernel P_R and δ quantifies the magnitude of non-stationarity.

Closed-form expressions and bounds.. Consequently, we obtain the following expressions for σ_f^2 and bounds for ρ :

$$\sigma_f^2 = \pi(1)\pi(0) + \frac{\pi(1)}{\pi(0)} \cdot \frac{1 - \gamma_{M,R}}{\gamma_{M,R}},$$

and

$$\left| \rho - \frac{1}{|D|\pi(1)} \right| \leq L \cdot \delta,$$

where L depends on the transition kernel P_R .

In particular, the dominating term in σ_f^2 satisfies

$$\sigma_f^2 \approx \frac{\pi(1)}{\gamma_{M,R}},$$

and the bounds on ρ can be equivalently written as

$$\frac{1}{|D|\pi(1)} - L\delta \leq \rho \leq \frac{1}{|D|\pi(1)} + L\delta.$$

Variance comparison across three regimes.. Let the benchmark term be $\frac{1}{|D|\pi(1)}$ with uncertainty radius $L\delta$. Then, the comparison between σ_f^2 and ρ yields three regimes. First, if

$$\frac{\pi(1)}{\gamma_{M,R}} > \frac{1}{|D|\pi(1)} + L\delta \quad (\text{i.e., } \sigma_f^2 \geq \rho),$$

then splitting yields a *slight increase* in variance.

Second, if

$$\frac{\pi(1)}{\gamma_{M,R}} < \frac{1}{|D|\pi(1)} - L\delta,$$

then the splitting algorithm has *smaller variance*, demonstrating its advantage in the low-bug-rate regime.

Third, if

$$\frac{1}{|D|\pi(1)} - L\delta < \frac{\pi(1)}{\gamma_{M,R}} < \frac{1}{|D|\pi(1)} + L\delta,$$

then σ_f^2 and ρ are *comparable*. In this indeterminate middle regime, we cannot definitively claim which variance is larger, and neither splitting nor non-splitting exhibits a significant variance advantage based on these bounds alone. The uncertainty term $L\delta$ (driven by P_R) therefore creates a middle region in which variance ordering is not identifiable from bounds alone.

Sample complexity advantage.. Regardless of the variance ordering, splitting improves sample complexity. Specifically, the non-splitting method has sample complexity $2T$, whereas the splitting method has sample complexity $2T - \tau$. Since $2T - \tau < 2T$, splitting has *lower sample complexity* than non-splitting. In other words, even when variance benefits are unclear (i.e., in the indeterminate middle regime), splitting still provides a consistent improvement through reduced sample complexity. A more rigorous statement requires computing the overall work efficiency by jointly accounting for variance and sample complexity.

We stress that, although our analysis is built on an ergodicity assumption, this condition plays an important

practical role: it rules out poorly performing fuzzers whose mutation processes can become trapped in restricted regions of the input space. Notably, our empirical results align closely with the theoretical predictions even when this assumption is not enforced. Similar assumptions are standard in the Markov chain literature. While it is certainly of interest to derive stronger results that relax ergodicity—and we hope this work motivates such future efforts—any rigorous analysis must begin with a baseline. In this sense, our work constitutes a first step, providing the initial analytical framework for studying Markov chain behavior in the context of fuzzing.

We further assume that the Markov chain is time-homogeneous, i.e., its transition matrix does not vary over time. This assumption is satisfied by most existing fuzzers, since their mutation rules are fixed and implemented as static code, which uniquely determines the transition matrix.

Finally, although our analysis focuses on the case of a single input seed, scenarios with multiple seeds can be accommodated naturally. If multiple seeds are executed in parallel, the process is equivalent to running multiple independent trials, which is already covered by our framework. If, instead, seeds are fixed in advance and cycled over time, this can be modeled by augmenting the transition matrix to include deterministic transitions to new seeds, followed by a restart under the original transition dynamics—effectively inducing a regeneration structure in the Markov chain.

Consider the case where D is a grid of integers in the set $[0, 9] \times [0, 9]$ in two-dimensional space. Suppose the set of bugs forms the sub-grid $[3, 5] \times [3, 5]$, i.e., a compact and convex subset of the input space. In this case, the path will split whenever it enters $[3, 5] \times [3, 5]$.

Assume the fuzzer is AFL, in that it randomly selects a location between 1 and 2 and modifies the value at the selected location with a random value. Notably, AFL is reversible which allows us to utilize the conclusion above. Without splitting, then we can write down the transition matrix as

$$P(x, y) = \frac{1}{10} 1_{x_1=y_1, x_2=y_2} + \left(\frac{1}{20}\right) 1_{x_1=y_1, x_2 \neq y_2} + \left(\frac{1}{20}\right) 1_{x_1 \neq y_1, x_2=y_2}$$

Step 1: define a state-dependent autocovariance weight.. Let $f = \mathbf{1}_B$ and $p := \pi(B)$. Define the centered function $g := f - p$. For a Markov chain with kernel P , define the state-dependent (signed) deviation

$$\Delta_\ell(x) := (P^\ell f)(x) - p = \mathbb{P}_x(X_\ell \in B) - \pi(B),$$

and define the local weight

$$c(x) := g(x)^2 + 2 \sum_{\ell \geq 1} |g(x) \Delta_\ell(x)|.$$

This quantity is the natural refinement of a global “integrated autocovariance”: it measures how strongly the event $X_t \in B$ remains informative about future membership in B , starting

from state x . In particular, for any time t , one may take $c_t := c(X_t)$ in the bound

$$\text{Var}(\hat{\mu}_T) \leq \frac{1}{T^2} \sum_{t=1}^T \frac{c(X_t)}{k_t},$$

which makes clear that times/states with larger $c(\cdot)$ contribute more and are precisely the ones you want larger k_t on.

Step 2: compute π and $p = \pi(B)$ for the grid AFL kernel.. Let us consider

$$p = \pi(B) = \frac{|B|}{|D|} = \frac{9}{100}.$$

Moreover,

$$g(x) = \begin{cases} 1-p, & x \in B, \\ -p, & x \notin B. \end{cases}$$

Step 3: compute $\Delta_\ell(x) = \mathbb{P}_x(X_\ell \in B) - p$ explicitly.. Let $m := |[3, 5]| = 3$, so $m/10 = 3/10$. For any initial state $x = (x_1, x_2)$, define

$$a_i(x) := \mathbf{1}\{x_i \in [3, 5]\}, \quad i = 1, 2.$$

Key observation for this kernel: at each step you resample exactly one coordinate uniformly in $\{0, \dots, 9\}$. Hence, after ℓ steps, a given coordinate i is *never refreshed* with probability $(1/2)^\ell$, and is refreshed at least once with probability $1 - (1/2)^\ell$. Using a partition on whether one coordinate is never refreshed (which occurs iff all ℓ updates chose the other coordinate), one obtains for every $\ell \geq 1$:

$$\begin{aligned} \mathbb{P}_x(X_\ell \in B) &= \underbrace{\left(1 - 2^{1-\ell}\right) \left(\frac{m}{10}\right)^2}_{\text{both coordinates refreshed at least once}} + \\ &\quad \underbrace{2^{-\ell} \left(\frac{m}{10}\right) (a_1(x) + a_2(x))}_{\text{exactly one coordinate never refreshed}}. \end{aligned}$$

Since $p = (m/10)^2$, this implies the exact deviation formula

$$\Delta_\ell(x) = \mathbb{P}_x(X_\ell \in B) - p = 2^{-\ell} \left(\frac{m}{10}\right) \left(a_1(x) + a_2(x) - 2\frac{m}{10}\right).$$

Specialization to the clustered-bug case $x \in B$.. If $x \in B$, then $a_1(x) = a_2(x) = 1$, so for all $\ell \geq 1$,

$$\Delta_\ell(x) = 2^{-\ell} \left(\frac{m}{10}\right) \left(2 - 2\frac{m}{10}\right) = 2^{1-\ell} \left(\frac{m}{10}\right) \left(1 - \frac{m}{10}\right).$$

For $m = 3$, this becomes

$$\Delta_\ell(x) = 0.42 \cdot 2^{-\ell}, \quad x \in B.$$

This is an explicit *persistence/return characterization*: starting inside the bug cluster, the excess probability of still being in B after ℓ steps decays exactly as $2^{-\ell}$, with amplitude $(m/10)(1 - m/10)$ controlled by the cluster width m . (Heuristically, $2^{-\ell}$ is the probability that the update schedule has failed to “refresh away” at least one coordinate.)

Step 4: compute $c(x)$ explicitly for $x \in B$.. For $x \in B$, $g(x) = 1 - p$ and $\Delta_\ell(x) \geq 0$, so the absolute values drop:

$$c(x) = (1-p)^2 + 2(1-p) \sum_{\ell \geq 1} \Delta_\ell(x), \quad x \in B.$$

Using $\sum_{\ell \geq 1} 2^{-\ell} = 1$ and the formula above, for $x \in B$

$$\sum_{\ell \geq 1} \Delta_\ell(x) = \sum_{\ell \geq 1} 2^{1-\ell} \left(\frac{m}{10}\right) \left(1 - \frac{m}{10}\right) = 2 \left(\frac{m}{10}\right) \left(1 - \frac{m}{10}\right),$$

hence

$$c(x) = (1-p)^2 + 4(1-p) \left(\frac{m}{10}\right) \left(1 - \frac{m}{10}\right), \quad x \in B.$$

Plugging $m = 3$ and $p = 9/100$ yields for $x \in B$,

$$\begin{aligned} c(x) &= (0.91)^2 + 4(0.91)(0.3)(0.7) \\ &= 0.8281 + 0.7644 = 1.5925. \end{aligned}$$

This implies that, for $x \in B$, the contribution term satisfies $\frac{c(x)}{k_t} = \frac{1.5925}{2} < 1$, which demonstrates the advantage of splitting at the bug-triggering point.

5.4. Measure on Splitting Performance. Consistent with the existing work on splitting, we assess the effectiveness of the proposed splitting strategy τ by comparing the variance of the resulting estimator to that under the optimal strategy τ^* . Specifically, we consider the relative efficiency

$$R_v = \frac{\mathcal{D}(\tau^*)}{\mathcal{D}(\tau)},$$

where

$$\mathcal{D}(\tau) = \int \tau^{-1}(x) \text{Var}(\hat{P}_R(T)) \pi^2(x) dx,$$

and τ is defined by

$$\tau(x) = \frac{\beta(x)\pi(x)}{\int \beta(u)\pi(u) du}.$$

Here, $\beta(x)$ denotes the splitting kernel associated with state x .

Under our splitting algorithm, natural choices include $\beta(x) = \mathbb{P}(R(x) \in B)$ or $\beta(x) = \mathbf{1}\{R(x) \in B\}$. These choices lead to different implications for the efficiency of the resulting fuzzers.

Theorem 6 (efficiency). *Let us assume that there exists δ , such that $1 \geq \delta_1 \geq P(R(x) \in B) \geq \delta > 0$ for any x . Let us also assume that $P(R(x) \in B)$ is computable. Let us assume that $\beta(x) = P(R(x) \in B)$. Then we have*

$$R_v \geq V,$$

where $V = \frac{\delta}{\delta_1}$.

Theorem 7 (efficiency). *Let us assume that there exists δ , such that $P(R(x) \in B) > \delta$ for any x . Let us also assume*

that $\omega(x) = 1_{R(x) \in B}$ and $\beta(x) = \min(\omega(x), a)$. Then we have

$$R_v \geq V$$

where $V = \max_a \min_x a \cdot P(R(x) \in B) \cdot P(R(x) \notin B) + P(R(x) \in B)^2$.

This implies that the resulting variance is at most $\frac{1}{V}$ times the optimal variance, where the optimum corresponds to the best achievable (but generally unknown) splitting strategy. Notably, the bound holds under mild conditions on the fuzzer, such as requiring the bug-reaching probability from any input to be uniformly bounded away from 0.

In this section, we provide a theoretical study of the proposed performance estimators—both asymptotically and in finite time—and, consequently, of the newly introduced evaluation measures. We consider both asymptotic and finite-time regimes with respect to the fuzzing horizon T and the number of replications M . The motivation for explicitly incorporating M is as follows.

For each replication $m \in \{1, 2, \dots, M\}$, the estimator

$$\tilde{P}_R^{(m)}(T) := \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{X_t^m \in B\}}$$

is computed from an independent simulation run (labeled by m). Since these replications are generated independently and are initialized from the same distribution, the collection $\{\tilde{P}_R^{(m)}(T)\}_{m=1}^M$ is (conditionally) i.i.d. across m . This experimental design aligns with the constrained super-Markov chain perspective formalized in [22], which advocates using many short chains rather than a single long chain, which is a recommendation that fits naturally in fuzz testing. Under this framework, the variability of $\tilde{P}_R^{(m)}(T)$ is governed by two factors: the convergence and mixing behavior of a single Markov chain as T increases, and the reduction in Monte Carlo error obtained by averaging across replications as M increases. Intuitively, an appropriate choice of M can reduce the required horizon T , enabling more efficient fuzzing in parallel computing environments.

We present the asymptotic analytical results in Section F, the finite-time results in Section G, and the corresponding evaluation framework in Section H.

6. Asymptotic analyses

In what follows, we analyze the asymptotic behavior of the estimator $\tilde{S}_R(T)$ and the corresponding variance of $\tilde{P}_R(T)$, which defines the evaluation measure $V_R(T)$, as well as the measures $A_R(T)$, $B_R(T)$, and $M_R(T)$.

Lemma 8. *For any ergodic Markov chain P_R , we obtain $\lim_{T \rightarrow \infty} \tilde{P}_R(T) = P_R = \pi(1)$.*

Proof. Let $\tau^+(B, m)$ denote the number of steps until the first bug is detected in the m th trial. By definition, the random variables

$$\tau^+(B, 1), \tau^+(B, 2), \dots, \tau^+(B, M)$$

are i.i.d. Hence, by the law of large numbers,

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{m=1}^M \tau^+(B, m) = \mathbb{E}[\tau^+(B, 1)] = \mathbb{E}[\tau^+(B)].$$

On the other hand, $\tau^+(B)$ is the first hitting time of the bug set B , i.e.,

$$\mathbb{E}[\tau^+(B)] = \mathbb{E}[\min\{t \geq 0 : X(t) \in B\}] = \mathbb{E}[\min\{t \geq 0 : N_t = 1\}].$$

Under the stationary distribution, the probability of being in B (within the relevant domain D) is $\pi(1) = \Pr(X \in B \mid X \in D)$. Therefore, the expected waiting time until the first visit to B satisfies

$$\mathbb{E}[\tau^+(B)] = \frac{1}{\Pr(X \in D, X \in B)} = \frac{1}{\pi(1)}.$$

Equivalently,

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{m=1}^M \tau^+(B, m) = \frac{1}{\pi(1)}.$$

Next, define the successive inter-arrival times between bug detections. Let

$$\tau^{+,1}(B) := \min\{t \geq \tau^+(B) : N_t = 2\},$$

and more generally, for any $n \geq 1$,

$$\tau^{+,n}(B) := \min\{t \geq \tau^{+,n-1}(B) : N_t = n\}.$$

By the Markov property and time-homogeneity of the transition kernel, the increments

$$\tau^+(B), \tau^{+,1}(B), \dots, \tau^{+,n}(B)$$

are i.i.d., each with mean $\mathbb{E}[\tau^+(B)]$.

By definition of N_T , we have the bracketing relation

$$\sum_{n=1}^{N_T} \tau^{+,n}(B) < T \leq \sum_{n=1}^{N_T+1} \tau^{+,n}(B),$$

which implies

$$\frac{1}{N_T} \sum_{n=1}^{N_T} \tau^{+,n}(B) < \frac{T}{N_T} \leq \frac{1}{N_T} \sum_{n=1}^{N_T+1} \tau^{+,n}(B).$$

Applying the law of large numbers to both averages yields

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{N_T} \sum_{n=1}^{N_T} \tau^{+,n}(B) &= \mathbb{E}[\tau^+(B)], \\ \lim_{T \rightarrow \infty} \frac{1}{N_T} \sum_{n=1}^{N_T+1} \tau^{+,n}(B) &= \mathbb{E}[\tau^+(B)]. \end{aligned}$$

By squeezing, we conclude that

$$\lim_{T \rightarrow \infty} \frac{T}{N_T} = \mathbb{E}[\tau^+(B)],$$

or equivalently,

$$\lim_{T \rightarrow \infty} \frac{N_T}{T} = \frac{1}{\mathbb{E}[\tau^+(B)]} = \Pr(X \in B \mid X \in D),$$

which establishes the desired claim and completes the proof. $\square$

Theorem 9 (Asymptotic property of sample variance (as $T \rightarrow \infty$)). *Let us assume that the transition matrix is ergodic. Then we have as T goes to ∞ , the sample variance reads as*

$$\lim_{T \rightarrow \infty} \tilde{S}_R(T) = 0.$$

Theorem 10 (Asymptotic property of sample variance (as $M \rightarrow \infty$)). *Let us assume that the transition matrix is ergodic. Then we have as M goes to ∞ , the sample variance reads as*

$$\lim_{M \rightarrow \infty} \tilde{S}_R(T) = S_R(T).$$

Theorem 11 (Asymptotic property of population variance (as $T \rightarrow \infty$)). *Let us assume that the transition matrix is ergodic. Then we have that as T goes to ∞ , the population variance reads as*

$$\lim_{T \rightarrow \infty} V_R(T) = 0.$$

Theorem 12 (Asymptotic property of population variance (as $M \rightarrow \infty$)). *Let us assume that the transition matrix is ergodic. Then we have that as M goes to ∞ , the population variance reads as*

$$\lim_{M \rightarrow \infty} V_R(T) = 0.$$

This implies that, with high probability, the variances of the estimator $\tilde{P}_R(T)$ —including both the sample variance and the population variance—approach zero as time goes to infinity. For any fixed T , as $M \rightarrow \infty$, both the sample variance $S_R(T)$ and the population variance $V_R(T)$ converge to the finite-time performance measure $S_R(T)$. This holds for every fuzzer $R \in \mathcal{R}$.

Theorem 13 (Asymptotic property of $M_R(T)$ (as $T \rightarrow \infty$)). *Let us assume that the transition matrix is ergodic. Then we have that as T goes to ∞ ,*

$$\lim_{T \rightarrow \infty} M_R(T) = 0.$$

Consequently, we conclude that the mean squared error of $\tilde{P}_R(T)$ also converges to zero asymptotically with respect to T .

Likewise, we consider the asymptotic behavior of $M_R(T)$ when M goes to ∞ .

Theorem 14 (Asymptotic property of $M_R(T)$ (as $M \rightarrow \infty$)). *Let us assume that the transition matrix is ergodic. Then we have that as M goes to ∞ ,*

$$\lim_{M \rightarrow \infty} M_R(T) = (P_R(T) - \pi(1))^2.$$

Thus far, convergence with respect to the fuzzing duration T implies convergence to zero. Naturally, we are also interested in asymptotic convergence toward a non-zero value, which characterizes the rate of convergence. This is particularly important for establishing the convergence

of $A_R(T)$ and $B_R(T)$. To this end, we present the central limit theorem for Markov chains [31], which enables us to characterize the non-zero asymptotic behavior of the variance. The formal statement is as follows.

Theorem 15 (Central Limit Theorem [31]). *Let $\{X_t\}_{t \geq 0}$ be a Markov chain with stationary distribution π , and let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a function such that $\mathbb{E}_\pi[f(X_0)^2] < \infty$. Assume the Markov chain is geometrically ergodic, i.e., there exist constants $C < \infty$ and $0 < \lambda_{R,2} < 1$ such that for all $x \in \mathcal{X}$ and $t \geq 0$,*

$$\|P^t(x, \cdot) - \pi(\cdot)\|_{TV} \leq C\lambda_{R,2}^t.$$

Define the sample mean:

$$\bar{f}_T = \frac{1}{T} \sum_{t=1}^T f(X_t).$$

Then the Central Limit Theorem holds:

$$\sqrt{T} (\bar{f}_T - \mathbb{E}_\pi[f]) \xrightarrow{d} \mathcal{N}(0, \sigma_f^2),$$

where

$$\begin{aligned} \sigma_f^2 &= \lim_{T \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T f(X_t) \right) \\ &= \text{Var}_\pi(f) + 2 \sum_{k=1}^{\infty} \text{Cov}_\pi(f(X_0), f(X_k)). \end{aligned}$$

In our context, $\bar{f}_T$ corresponds to $\tilde{P}_R(T)$ and $f = \mathbf{1}_{\{X \in B\}}$. We also assume that the Markov chain is geometrically ergodic. As a result, we leverage the central limit theorem to characterize robustness and to estimate P_R .

The above theorem establishes that $T \cdot V_R(T)$ and $T \cdot S_R(T)$ converge to $\sigma_f^2 = \sigma^2$, as defined previously. Equivalently, $\lim_{T \rightarrow \infty} A_R(T) = 0$, which in turn implies the following asymptotic result for $A_R(T)$.

Corollary 1 (Asymptotic property of $A_R(T)$). *Let us assume that the transition matrix is ergodic. Then, as T goes to ∞ , we have*

$$\lim_{T \rightarrow \infty} A_R(T) = 0.$$

Meanwhile, based on the aforementioned result for $V_R(T)$, we obtain the following asymptotic property of $A_R(T)$ with respect to M .

Corollary 2 (Asymptotic property of $A_R(T)$). *Let us assume that the transition matrix is ergodic. Then, as M goes to ∞ , we have*

$$\lim_{M \rightarrow \infty} A_R(T) = |T \cdot S_R(T) - \sigma^2|.$$

For the deviation of $\tilde{S}_R(T)$ from σ^2 , we obtain the following asymptotic convergence result for $B_R(T)$.

Corollary 3 (Asymptotic property of $B_R(T)$). *Let us assume that the transition matrix is ergodic. Then, as T goes to ∞ , we have*

$$\lim_{T \rightarrow \infty} B_R(T) = \left| \lim_{T \rightarrow \infty} T \cdot V_R(T) - \sigma^2 \right| = 0.$$

Additionally, the convergence result for $S_R(T)$ implies the following asymptotic property of $B_R(T)$ with respect to M .

Corollary 4 (Asymptotic property of $B_R(T)$). *Let us assume that the transition matrix is ergodic. Then, as M goes to ∞ , we have*

$$\lim_{M \rightarrow \infty} B_R(T) = |T \cdot S_R(T) - \sigma^2|.$$

Essentially, the convergence of $A_R(T)$, $B_R(T)$, and $M_R(T)$ is established via the corresponding results for $S_R(T)$ and $V_R(T)$, which substantially simplifies the analyses. We carry this observation forward in our finite-time analyses.

Next, it remains essential to understand the finite-time behavior of the estimators and to examine how closely they approximate their steady-state counterparts. This understanding is key to validating our methodology for assessing variability in the bug detection rate, constructing confidence intervals based on the central limit theorem, and evaluating the robustness of bug detection rates across different fuzzers.

7. Finite-time Analyses

In this section, we examine the finite-time behavior of the proposed performance estimator and evaluation measures. Specifically, we characterize the finite-time properties of variances $\tilde{S}_R(T)$ and $V_R(T)$. Building on these results, we extend the finite-time analysis to the error metrics $A_R(T)$ and $B_R(T)$, as well as the mean squared error $M_R(T)$.

More precisely, we begin by studying how the sample variance $\tilde{S}_R(T)$ and the population variance $V_R(T)$ converge to σ^2 , which allows us to derive corresponding upper bounds on $A_R(T)$ and $B_R(T)$. We then establish finite-time upper bounds on $\tilde{S}_R(T)$ and $V_R(T)$, which in turn enable us to obtain an upper bound on $M_R(T)$.

Theorem 16 (Convergence of Sample Variance). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic. Let us assume that the simulation runs for T time steps. Then We have with probability at least $1 - 2 \exp\{-\frac{M}{S^2}\} = 1 - 2 \exp\{-MT\}$ that*

$$\begin{aligned} & |T \cdot \tilde{S}_R(T) - \sigma^2| \\ & \leq \frac{1}{M} + \left(\frac{1}{4} + \frac{1}{16\pi(1)^2}\right) \cdot \frac{(1 - \gamma_{M,R})(1 + (1 - \gamma_{M,R}))}{(1 - (1 - \gamma_{M,R}))^2} \\ & \quad + \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{\gamma_{M,R}^2} \\ & \quad + 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}} \end{aligned}$$

where $\gamma_{M,R}$ denotes the spectral gap constant of the Markov chain. In the non-reversible case, $\gamma_{M,R}$ is defined as the spectral gap of a suitable reversibilization of the chain. Specifically, let $P_{M,R}$ be a reversible kernel with respect

to π_R obtained from P_R (i.e., a reversibilization). We then define

$$\gamma_{M,R} := 1 - \lambda_2(P_{M,R}).$$

Theorem 17 (Convergence of Population Variance). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic. Let us assume that the simulation runs for T time steps. Then We have that*

$$\begin{aligned} & |T \cdot V_R(T) - \sigma^2| \\ & = |Var(\sqrt{T} \tilde{P}_R(T)) - \sigma^2| \\ & \leq \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} + \\ & \quad 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}} \end{aligned}$$

Based on the above convergence results, we derive the following straight-forward corollary on the upper bound of $A_R(T)$ and $B_R(T)$.

Corollary 5 (Upper bounds on $A_R(T)$ and $B_R(T)$). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic. Let us assume that the simulation runs for T time steps. Then We have that*

$$\begin{aligned} & A_R(T) \\ & \leq \frac{1}{M} + \left(\frac{1}{4} + \frac{1}{16\pi(1)^2}\right) \cdot \frac{(1 - \gamma_{M,R})(1 + (1 - \gamma_{M,R}))}{(1 - (1 - \gamma_{M,R}))^2} \\ & \quad + \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{\gamma_{M,R}^2} \\ & \quad + 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}} \end{aligned}$$

and

$$\begin{aligned} & B_R(T) \\ & \leq \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} + \\ & \quad 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}}. \end{aligned}$$

Additionally, we establish finite-time upper bounds on $\tilde{S}_R(T)$ and $V_R(T)$ that hold for all T . These results yield conservative guarantees when $\tilde{P}_R(T)$ must be computed using only empirical estimators. We present bounds for both reversible and non-reversible Markov chains, since the resulting expressions depend on whether the chain is reversible; moreover, the analytical techniques required in the two cases differ substantially.

We begin with the reversible setting, where the transition kernel $P_R(\cdot, \cdot)$ satisfies the *detailed balance* (reversibility) condition: there exists a stationary distribution π_R such that, for all states $x, y \in \mathcal{X}$,

$$\pi_R(x) P_R(x, y) = \pi_R(y) P_R(y, x).$$

Reversible Markov chains.

Theorem 18 (Sample variance). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic, reversible, and positive. Let us assume that the simulation runs for T time steps. Then We have*

$$\begin{aligned}\sqrt{\tilde{S}_R(T)} &\leq \sqrt{2 \frac{\text{Var}(\sum_{t=1}^T 1_{X_t^m \in B})}{M} + (\frac{4\delta_R}{T})^2} \\ &\leq \sqrt{2 \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2} + (\frac{4\delta_R}{T})^2}\end{aligned}$$

where K_R is defined as in [32].

Subsequently, we establish an upper bound on the population variance, stated in the following result for $V_R(T)$.

Theorem 19 (Population variance). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic, reversible, and positive. Let us assume that the simulation runs for T time steps. Then we obtain that*

$$\begin{aligned}V_R(T) &\leq \min(2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} + \\ &2(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}})^2, \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2}).\end{aligned}$$

where K_R is defined as in [32].

The above results establish robustness guarantees for the fuzzers, but they apply only to reversible Markov chains, which can be restrictive in practice. We therefore turn to general non-reversible chains and derive corresponding finite-time upper bounds on the sample and population variances.

Non-reversible Markov chains. For general geometrically ergodic Markov chains (without assuming reversibility), prior work derives variance bounds by relating the chain to a Metropolis-Hastings (M-H) *reversibilization* and its spectral gap. In what follows, we adopt the framework of [4] and begin by defining the M-H spectral gap under M-H *reversibilization*.

Definition 1 (M-H spectral gap). *Let us assume that M_1 and M_2 are the first and second Metropolis-Hastings (M-H) Markov chains associated with P_R . Then, the M-H spectral gap of P_R is defined as follows:*

$$\gamma_R^{MH} = 1 - \sup_{n \in \mathbb{N}} \{L(M_2(P_R^n))^{\frac{1}{n}}, U(M_1(P_R^n))^{\frac{1}{n}}\}$$

where L and U represents the operators on a transition matrix P

$$\begin{aligned}L(P) &= \inf_{f \in L_0^2(\pi), \|f\|_\pi = 1} \langle Pf, f \rangle_\pi \\ U(P) &= \sup_{f \in L_0^2(\pi), \|f\|_\pi = 1} \langle Pf, f \rangle_\pi\end{aligned}$$

Additionally, we define C and the complement of C as C^c as follows. Formally, we define a feasible region C as

$$C = \{n \in \mathbb{N} : L(M_2(P_R^n)) \leq 1, U(M_1(P_R^n)) \leq 1\}$$

and

$$C^c = \mathbb{N} \setminus C.$$

With these definitions and tools for the non-reversible setting in place, we proceed to establish upper bounds on both the sample variance and the true variance via Metropolis-Hastings reversibilization for general Markov chains. The formal statements are as follows.

Theorem 20 (Sample variance). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic, and positive. Let us assume that the simulation runs for T time steps. We have*

$$\tilde{S}_R(T) \leq \frac{2}{M} (\pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}}) + (\frac{4\delta_R}{T})^2).$$

where δ_R is the corresponding relaxation time, defined as $\frac{1}{\gamma_R^{MH}}$.

Subsequently, we establish the following upper bound on the population variance $V_R(T)$.

Theorem 21 (Population variance). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic, and positive. Let us assume that the simulation runs for T time steps. Then we obtain that*

$$V_R(T) \leq \pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}}).$$

Next, we turn to the finite-time analysis of the mean squared error for a fixed simulation horizon T . Specifically, we derive finite-time upper bounds on the MSE to compare worst-case performance across different fuzzers. As before, we present results for both reversible and non-reversible Markov chains, reflecting the distinct structural properties of their transition kernels.

We emphasize that, given our bounds on $V_R(T)$ and the bias term, an upper bound on the mean squared error $M_R(T)$ follows directly. We begin with the reversible case in Theorem 22 and then extend the analysis to general non-reversible chains in Theorem 23.

Theorem 22 (Mean Square Error). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic, reversible, and positive. Let us assume that the simulation runs for T time steps. Then we derive that*

$$\begin{aligned}M_R(T) &\leq \min(2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} \\ &+ 2(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}})^2, \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2}) \\ &+ (\min(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}))^2.\end{aligned}$$

where C_R, α_R, K_R are defined as: $C_R = \alpha^{-r}$, $\alpha_R = \theta^{\frac{1}{r}}$ are constants associated with the transition matrix P_R^X , with $\theta = 1 - \min_{x_1, x_2} (\frac{P_R^X(x_1, x_2)}{\pi(x_2)})^r$, $r = \min_t (P_R^X)^t(x_1, x_2) > 0$

for any x_1, x_2 ; $(P_R^X)^t$ denotes P_R^X to the power of t ; K_R is defined as in Theorem 19 following [32].

Theorem 23 (Mean Square Error). *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic, and positive. Let us assume that the simulation runs for T time steps. We hence have*

$$M_R(T) \leq \frac{\pi(1)(1 - \pi(1)(|C|^c + \frac{2}{\gamma_R^{MH}}))}{MT} + (\min(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}))^2.$$

8. Evaluation Results

8.1. $V_R(T)$ -based Evaluation. We next present the evaluation based on $V_R(T)$.

Theorem 24. *For each fuzzer $R \in \mathcal{R}$, define an upper bound $U_R(T)$ on $V_R(T)$ as follows.*

- 1) **Reversible Markov chains.** Define $U_R^{\text{rev}}(T) := \min\left(2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} + 2\left(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}}\right)^2, \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2}\right)$, where $K_R \leq 1 + \frac{2\lambda_{R,2}}{T(1 - \lambda_{R,2})^2}$.
- 2) **Non-reversible Markov chains.** Define $U_R^{\text{nonrev}}(T) := \pi(1)(1 - \pi(1))\left(|C|^c + \frac{2}{\gamma_R^{MH}}\right)$.

Selecting any $\hat{R} \in \arg \min_{R \in \mathcal{R}} U_R^{\text{rev}}(T)/U_R^{\text{nonrev}}(T)$ yields a fuzzer that minimizes this upper bound on $V_R(T)$, and thus identifies the best fuzzer under this criterion.

8.2. $S_R(T)$ -based Evaluation.

Theorem 25. *Let T be the simulation horizon and M the number of trials. For each fuzzer $R \in \mathcal{R}$, define an upper bound $U_R(T)$ on the deviation $\tilde{S}_R(T)$ as follows.*

- 1) **Reversible Markov chains.** Define $U_R^{\text{rev}}(T) := \frac{2}{M} \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2} + \left(\frac{4\delta_R}{T}\right)^2$, where $K_R \leq 1 + \frac{2\lambda_{R,2}}{T(1 - \lambda_{R,2})^2}$.
- 2) **Non-reversible Markov chains.** Define $U_R^{\text{nonrev}}(T) := \frac{2}{M} \left(\pi(1)(1 - \pi(1))\left(|C|^c + \frac{2}{\gamma_R^{MH}}\right) + (4\delta_R)^2\right)$.

Selecting any $\hat{R} \in \arg \min_{R \in \mathcal{R}} U_R^{\text{rev}}(T)/U_R^{\text{nonrev}}(T)$ yields a fuzzer with the smallest deviation $\tilde{S}_R(T)$, and thus selects the best fuzzer under this criterion.

8.3. $M_R(T)$ -based Evaluation. Lastly, we turn to an evaluation based on $M_R(T)$, which incorporates both variance and bias information and is widely used in machine learning and statistics.

Theorem 26. *For each fuzzer $R \in \mathcal{R}$, define an upper bound $U_{M,R}(T)$ on $M_R(T)$ as follows.*

- 1) **Reversible Markov chains.** Define $U_{M,R}^{\text{rev}}(T) := \min\left(2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} + 2\left(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}}\right)^2, \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2}\right) +$

$$\left(\min\left(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}\right)\right)^2, \text{ where } K_R \leq 1 + \frac{2\lambda_{R,2}}{T(1 - \lambda_{R,2})^2}.$$

2) **Non-reversible Markov chains.** Define $U_{M,R}^{\text{nonrev}}(T) := \frac{\pi(1)(1 - \pi(1))\left(|C|^c + \frac{2}{\gamma_R^{MH}}\right)}{MT} + \left(\min\left(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}\right)\right)^2.$

Selecting any $\hat{R} \in \arg \min_{R \in \mathcal{R}} U_{M,R}^{\text{rev}}(T)/U_{M,R}^{\text{nonrev}}(T)$ yields a fuzzer that minimizes this upper bound on $M_R(T)$, and thus selects the best fuzzer under this criterion.

9. Section 5.1

Proof of Theorem 9. We note the following key observation (by definition):

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B} = P_R(T).$$

Consequently, we obtain

$$\begin{aligned} \lim_{T \rightarrow \infty} \tilde{S}_R(T) &= \lim_{T \rightarrow \infty} \sqrt{\frac{1}{M} \sum_{m=1}^M \left(\tilde{P}_R(T) - \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B}\right)^2} \\ &= \lim_{T \rightarrow \infty} \sqrt{\frac{1}{M} \sum_{m=1}^M \left(P_R(T) - \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B}\right)^2} \\ &= \lim_{T \rightarrow \infty} \sqrt{\frac{1}{M} \sum_{m=1}^M (P_R(T) - P_R(T))^2} \\ &= 0, \end{aligned}$$

which completes the proof. $\square$

Proof of Theorem 10. By definition,

$$\tilde{S}_R(T) = \frac{1}{M} \sum_{m=1}^M \left(\tilde{P}_R(T) - \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B}\right)^2,$$

where $\frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B}$ are independently and identically distributed based on the MC simulation process. Then, by the strong law of large numbers, we obtain

$$\begin{aligned} \lim_{M \rightarrow \infty} \tilde{S}_R(T) &= E\left[\left(\tilde{P}_R(T) - \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B}\right)^2\right] \\ &= \text{Var}\left(\frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B}\right) = S_R(T), \end{aligned}$$

which concludes the proof. $\square$

Proof of Theorem 11. By definition, we obtain

$$\begin{aligned}
& \lim_{T \rightarrow \infty} V_R(T) \\
&= \lim_{T \rightarrow \infty} \sqrt{\frac{1}{M} \sum_{m=1}^M (E[\tilde{P}_R(T)] - \tilde{P}_R(T))^2} \\
&= \lim_{T \rightarrow \infty} \sqrt{\frac{1}{M} \sum_{m=1}^M (\pi(1) - \tilde{P}_R(T))^2} \\
&= \lim_{T \rightarrow \infty} \sqrt{\frac{1}{M} \sum_{m=1}^M (\pi(1) - \pi(1))^2} \\
&= 0,
\end{aligned}$$

which finishes the proof. $\square$

Proof of Theorem 12. Based on

$$V_R(T) = E[(\tilde{P}_R(T) - E[\tilde{P}_R(T)])^2],$$

we have that

$$\begin{aligned}
\lim_{M \rightarrow \infty} V_R(T) &= E\left[\lim_{M \rightarrow \infty} (\tilde{P}_R(T) - E[\tilde{P}_R(T)])^2\right] \\
&= E[(E[\tilde{P}_R(T)] - E[\tilde{P}_R(T)])^2] \\
&= 0,
\end{aligned}$$

which completes the proof. $\square$

Proof of Theorem 13. Again, by definition,

$$\lim_{T \rightarrow \infty} M_R(T) = \lim_{T \rightarrow \infty} V_R(T) + (E[D_R(T)])^2.$$

Meanwhile, by Theorem 11 and by Lemma 8,

$$\lim_{T \rightarrow \infty} V_R(T) = 0, \quad \lim_{T \rightarrow \infty} D_R(T) = 0,$$

and the claim follows by exchanging the limit and expectation. $\square$

Proof of Theorem 14. Notably,

$$\begin{aligned}
\lim_{M \rightarrow \infty} M_R(T) &= \lim_{M \rightarrow \infty} E\left(\tilde{P}_R(T) - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T)\right) \\
&= E[(P_R(T) - P_R)^2] \\
&= (P_R(T) - P_R)^2 = (P_R(T) - \pi(1))^2.
\end{aligned}$$

$\square$

10. Section 5.2

Proof. Theorem 16 Let us recall that $\rho = \lambda_{R,2}$ is the second largest eigenvalue of P_R . For example, in the two-state model, we derive that $\lambda_{R,2} = 1 - P_R(0, 1) - P_R(1, 0)$.

First, by definition, we arrive at

$$\begin{aligned}
& \text{Var}(TP_R(T)) \\
&= \frac{1}{M} \sum_{i=1}^T \sum_{j=1}^T \text{Cov}(1_{X_i^m \in B}, 1_{X_j^m \in B}) \\
&= \sum_{t=-(T-1)}^{(T-1)} (T-t)\gamma_t
\end{aligned}$$

where $\gamma_t = \text{Cov}(1_{X_0^m \in B}, 1_{X_t^m \in B})$.

Notably, $\gamma_t = \gamma_{-t}$, and $\gamma_0 = \text{Var}(1_{X_0^m \in B})$, we obtain

$$\begin{aligned}
& \text{Var}(\sqrt{T}P_R(T)) \\
&= \frac{1}{T} \text{Var}(TP_R(T)) \\
&= \gamma_0 + 2 \sum_{t=1}^{(T-1)} (1 - \frac{t}{T})\gamma_t
\end{aligned}$$

Meanwhile, by definition,

$$\sigma^2 = \gamma_0 + 2 \sum_{t=1}^{\infty} \gamma_t$$

Subsequently, we derive

$$\begin{aligned}
& \text{Var}(\sqrt{T}P_R(T)) \\
&= \sigma^2 - 2 \sum_{t=1}^{\infty} \gamma_t + 2 \sum_{t=1}^{(T-1)} (1 - \frac{t}{T})\gamma_t \\
&= \sigma^2 - 2 \sum_{t=1}^{T-1} \gamma_t - 2 \sum_{t=T}^{\infty} \gamma_t + 2 \sum_{t=1}^{(T-1)} (1 - \frac{t}{T})\gamma_t \\
&= \sigma^2 - 2 \sum_{t=1}^{(T-1)} (\frac{t}{T})\gamma_t - 2 \sum_{t=T}^{\infty} \gamma_t
\end{aligned}$$

It is worth noting that for a geometrically ergodic markov chain, we have

$$\begin{aligned}
& |2 \sum_{t=T}^{\infty} \gamma_t| \leq 2 \sum_{t=T}^{\infty} |\gamma_t| \\
&= 2 \sum_{t=T}^{\infty} |\text{Cov}(1_{X_0^m \in B}, 1_{X_t^m \in B})| \\
&\leq 2 \sum_{t=T}^{\infty} |P(X_t^m \in B, X_0^m \in B) - P(X_t^m \in B)P(X_0^m \in B)| \\
&= 2 \sum_{t=T}^{\infty} |(P(X_t^m \in B|X_0^m \in B) - P(X_t^m \in B))P(X_0^m \in B)|
\end{aligned}$$

Notably, we have the following decomposition

$$\begin{aligned}
& |(P(X_t^m \in B|X_0^m \in B) - P(X_t^m \in B))| \\
&\leq |(P(X_t^m \in B|X_0^m \in B) - \pi(1) + \pi(1) - P(X_t^m \in B))| \\
&\leq \frac{(1 - \gamma_{M,R})^T}{4\pi(1)} + \frac{(1 - \gamma_{M,R})^T}{4 \max\{\pi(0), \pi(1)\}} \\
&\leq \frac{(1 - \gamma_{M,R})^T}{2 \max\{\pi(0), \pi(1)\}}
\end{aligned}$$

This immediately gives us that

$$\begin{aligned}
& |2 \sum_{t=T}^{\infty} \gamma_t| \\
& \leq 2 \sum_{t=T}^{\infty} |(P(X_t^m \in B | X_0^m \in B) - P(X_t^m \in B))P(X_0^m \in B)| \\
& \leq 2\pi(1) \sum_{t=T}^{\infty} \frac{(1 - \gamma_{M,R})^T}{2 \max\{\pi(0), \pi(1)\}} \\
& = 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}}
\end{aligned}$$

Consequently, we obtain

$$\begin{aligned}
|Var(\sqrt{T}P_R(T)) - \sigma^2| & \leq \left| \frac{1}{T} \sum_{t=1}^{\infty} t \cdot \gamma_t \right| \\
& + 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}}.
\end{aligned}$$

where $\gamma_t = Cov(1_{X_0^m \in B}, 1_{X_t^m \in B})$

More precisely, based on the above derivation, we have that for a geometrically ergodic Markov chain, we derive that

$$\gamma_t = \frac{\pi(1)(1 - \gamma_{M,R})^T}{2 \max\{\pi(0), \pi(1)\}}$$

which immediately implies that

$$\begin{aligned}
& \frac{2}{T} \sum_{t=1}^{\infty} t \cdot \gamma_t \\
& = \frac{2}{T} \sum_{t=1}^{\infty} t \cdot \frac{\pi(1)(1 - \gamma_{M,R})^T}{2 \max\{\pi(0), \pi(1)\}} \\
& = \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2}
\end{aligned}$$

Putting them together, we derive that

$$\begin{aligned}
|Var(\sqrt{T}P_R(T)) - \sigma^2| & \leq \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} \\
& + 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}}.
\end{aligned}$$

Next, we examine the difference between $Var(\sqrt{T}P_R(T))$ and $S_R(T)$. Notably,

$$E[S_R(T)] = Var_{\pi_0}(\sqrt{T}P_R(T))$$

Meanwhile,

$$\begin{aligned}
& |Var_{\pi_0}(\sqrt{T}P_R(T)) - Var(\sqrt{T}P_R(T))| \\
& \leq \frac{C_2}{T}
\end{aligned}$$

We next establish an upper bound on C_2 . From total variation distance estimates and standard covariance perturbation bounds (see), we use the following general result that

holds for bounded functions f and geometrically ergodic Markov chains, and is stated for example in Lemma 2.1 of [8]. Although their primary focus is on batch means estimators, this bound itself applies more broadly and is independent of the specific variance estimation method. Formally, we have

$$\begin{aligned}
& |\text{Cov}_{\pi_0}(f(X_i), f(X_j)) - \text{Cov}_{\pi}(f(X_i), f(X_j))| \\
& \leq \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) \cdot ((1 - \gamma_{M,R}))^{\max(i,j)}
\end{aligned}$$

where

$$f(x) = 1_{x \in B}.$$

This holds true by the following observation. Note that we assume that $i \leq j$ without loss of generalizability

$$\begin{aligned}
& |\text{Cov}_{\pi_0}(f(X_i), f(X_j)) - \text{Cov}_{\pi}(f(X_i), f(X_j))| \\
& \leq \left| P(X_i \in B, X_j \in B | X_0 \sim \pi_0) \right. \\
& \quad - P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_0 \sim \pi_0) \\
& \quad - P(X_i \in B, X_j \in B | X_0 \sim \pi) \\
& \quad \left. + P(X_i \in B | X_0 \sim \pi) \cdot P(X_j \in B | X_0 \sim \pi) \right| \\
& \leq \left| P(X_i \in B, X_j \in B | X_0 \sim \pi_0) \right. \\
& \quad - P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_0 \sim \pi_0) \\
& \quad - P(X_i \in B, X_j \in B | X_0 \sim \pi) \\
& \quad \left. + P(X_i \in B | X_0 \sim \pi) \cdot P(X_j \in B | X_0 \sim \pi) \right| \\
& \leq \left| P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_i \in B) \right. \\
& \quad - P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_0 \sim \pi_0) \\
& \quad - P(X_j \in B | X_i \in B) P(X_i \in B | X_0 \sim \pi) \\
& \quad \left. - P(X_i \in B | X_0 \sim \pi) \cdot P(X_j \in B | X_0 \sim \pi) \right| \\
& \leq P(X_j \in B | X_i \in B) \left| P(X_i \in B | X_0 \sim \pi_0) - P(X_i \in B | X_0 \sim \pi) \right| \\
& \quad + \left| P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_0 \sim \pi_0) \right. \\
& \quad \left. - P(X_i \in B | X_0 \sim \pi) \cdot P(X_j \in B | X_0 \sim \pi) \right| \\
& \leq P(X_j \in B | X_i \in B) \|\mu_i - \pi\|_{TV} \\
& \quad - P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_0 \sim \pi_0) \\
& \quad - P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_0 \sim \pi) \\
& \quad + P(X_i \in B | X_0 \sim \pi_0) \cdot P(X_j \in B | X_0 \sim \pi) \\
& \quad - P(X_i \in B | X_0 \sim \pi) \cdot P(X_j \in B | X_0 \sim \pi) \\
& \leq P(X_j \in B | X_i \in B) \|\mu_i - \pi\|_{TV} \\
& \quad - P(X_j \in B | X_0 \sim \pi) \|\mu_i - \pi\|_{TV} \\
& \quad + P(X_j \in B | X_0 \sim \pi) \|\mu_j - \pi\|_{TV} \\
& = (P(X_j \in B | X_i \in B) - P(X_j \in B | X_0 \sim \pi)) \|\mu_i - \pi\|_{TV} \\
& \quad + P(X_i \in B | X_0 \sim \pi) \|\mu_j - \pi\|_{TV} \\
& = \|\mu_{j-i} - \pi\|_{TV} \|\mu_i - \pi\|_{TV} \\
& \quad + P(X_i \in B | X_0 \sim \pi) \|\mu_j - \pi\|_{TV}
\end{aligned}$$

where $\mu_t = \mathcal{L}(X_t | X_0 \sim \pi_0)$.

Based on the total variation distance result:

$$\|\mu_t - \pi\|_{TV} = \frac{1}{2} |\mathbb{P}(X_t = 1) - \pi(1)| = \frac{1}{4\pi_1} (1 - \gamma_{M,R})^t,$$

where $\pi_1 = \frac{P_R(0,1)}{P_R(0,1)+P_R(1,0)}$ is the stationary probability of state 1.

Subsequently, we derive that

$$\begin{aligned} & |\text{Cov}_{\pi_0}(f(X_i), f(X_j)) - \text{Cov}_{\pi}(f(X_i), f(X_j))| \\ & \leq \frac{1}{4\pi_1} (1 - \gamma_{M,R})^{j-i} \frac{1}{4\pi_1} (1 - \gamma_{M,R})^i + \pi(1) \cdot \frac{1}{4\pi_1} (1 - \gamma_{M,R})^j \\ & \leq \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) (1 - \gamma_{M,R})^j \end{aligned}$$

Likewise, we can obtain the same result for $i > j$, and thus validate the claim.

Hence,

$$\begin{aligned} & |\text{Cov}_{\mu}(f(X_i), f(X_j)) - \text{Cov}_{\pi}(f(X_i), f(X_j))| \\ & \leq \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) \cdot (1 - \gamma_{M,R})^{\max(i,j)}. \end{aligned}$$

where

$$f(x) = 1_{x \in B}$$

This yields:

$$|C_2| \leq \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) (1 - \gamma_{M,R})^{\max(i,j)}.$$

We now compute the sum:

$$\begin{aligned} & \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} (1 - \gamma_{M,R})^{\max(i,j)} \\ & = 2 \sum_{k=1}^{\infty} k (1 - \gamma_{M,R})^k - \sum_{k=1}^{\infty} (1 - \gamma_{M,R})^k \\ & = 2 \cdot \frac{(1 - \gamma_{M,R})}{(1 - (1 - \gamma_{M,R}))^2} - \frac{(1 - \gamma_{M,R})}{1 - (1 - \gamma_{M,R})} \\ & = \frac{(1 - \gamma_{M,R})(1 + (1 - \gamma_{M,R}))}{(1 - (1 - \gamma_{M,R}))^2}. \end{aligned}$$

Therefore,

$$C_2 \leq \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) \cdot \frac{(1 - \gamma_{M,R})(1 + (1 - \gamma_{M,R}))}{(1 - (1 - \gamma_{M,R}))^2}$$

Likewise, if we start from $X_0 = 0$, then we replace $\pi(1)$ with $\pi(0)$, and everything follows the same logic. Henceforth, we skip the proof steps herein. As a result, we establish that

$$C_2 \leq \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) \cdot \frac{(1 - \gamma_{M,R})(1 + (1 - \gamma_{M,R}))}{(1 - (1 - \gamma_{M,R}))^2}.$$

This directly leads to

$$C_2 \leq \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) \cdot \frac{(1 - \gamma_{M,R})(1 + (1 - \gamma_{M,R}))}{(1 - (1 - \gamma_{M,R}))^2}.$$

Lastly, using the concentration inequality for $S_R(T)$ (Hoeffding inequality for bounded that holds true by the claim we obtain for the upper bound on $S_R(T)$ and independent random variables), we derive

$$\begin{aligned} & P(|S_R(T) - E[S_R(T)]| \geq \frac{1}{M}) \\ & \leq 2 \exp^{-\frac{M^2}{MS^2}} \\ & \leq 2 \exp^{-\frac{M}{S^2}} \end{aligned}$$

where S^2 is the upper bound we derive on $S_R(T)$.

Putting everything together, we obtain

$$\begin{aligned} & |S_R(T) - \sigma^2| \\ & = |S_R(T) - E[S_R(T)] + E[S_R(T)] - \text{Var}_{\pi_0}(\sqrt{T}P_R(T)) + \\ & \quad \text{Var}_{\pi_0}(\sqrt{T}P_R(T)) - \text{Var}(\sqrt{T}P_R(T)) + \\ & \quad \text{Var}(\sqrt{T}P_R(T)) - \sigma^2| \\ & \leq |S_R(T) - E[S_R(T)]| + |E[S_R(T)] - \text{Var}_{\pi_0}(\sqrt{T}P_R(T))| \\ & \quad + |\text{Var}_{\pi_0}(\sqrt{T}P_R(T)) - \text{Var}(\sqrt{T}P_R(T))| + \\ & \quad |\text{Var}(\sqrt{T}P_R(T)) - \sigma^2| \end{aligned}$$

With probability at least $1 - 2 \exp^{-\frac{M}{S^2}}$, we have that

$$\begin{aligned} & |S_R(T) - \sigma^2| \\ & \leq |S_R(T) - E[S_R(T)]| + |E[S_R(T)] - \text{Var}_{\pi_0}(\sqrt{T}P_R(T))| \\ & \quad + |\text{Var}_{\pi_0}(\sqrt{T}P_R(T)) - \text{Var}(\sqrt{T}P_R(T))| + \\ & \quad |\text{Var}(\sqrt{T}P_R(T)) - \sigma^2| \\ & \leq \frac{1}{M} + 0 + \frac{C_2}{T} + \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} + \\ & \quad 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}} \end{aligned}$$

Plugging in $C_2 = \frac{(P_R(0,1)+P_R(1,0))(1-\gamma_{M,R})(1+(1-\gamma_{M,R}))}{\min\{P_R(1,0), P_R(0,1)\}(1-(1-\gamma_{M,R}))^2}$, we arrive at

$$\begin{aligned} & |S_R(T) - \sigma^2| \\ & \leq \frac{1}{M} + \frac{C_2}{T} + \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} + \\ & \quad 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}} \\ & \leq \frac{1}{M} + \left(\frac{1}{4} + \frac{1}{16\pi_1^2}\right) \cdot \frac{(1 - \gamma_{M,R})(1 + (1 - \gamma_{M,R}))}{(1 - (1 - \gamma_{M,R}))^2} + \\ & \quad \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} + \\ & \quad 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}} \end{aligned}$$

Based on Theorem 16, we obtain that

$$S^2 \leq O\left(\frac{1}{T}\right)$$

and subsequently, we completes the proof steps.

Proof. Theorem 17 By definition, again, we have

$$|Var(\sqrt{T}P_R(T)) - \sigma^2| \leq \left| \frac{1}{T} \sum_{t=1}^{\infty} t \cdot \gamma_t \right| + 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}}$$

where $\gamma_t = Cov(1_{X_t^m \in B}, 1_{X_t^m \in B})$

More precisely, for a geometrically ergodic Markov chain, we derive that

$$\gamma_t = \frac{\pi(1)(1 - \gamma_{M,R})^T}{2 \max\{\pi(0), \pi(1)\}}$$

and subsequently

$$\begin{aligned} & \frac{2}{T} \sum_{t=1}^{\infty} t \cdot \gamma_t \\ &= \frac{2}{T} \sum_{t=1}^{\infty} t \cdot \frac{\pi(1)(1 - \gamma_{M,R})^T}{2 \max\{\pi(0), \pi(1)\}} \\ &= \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} \end{aligned}$$

Putting them together, we derive that

$$\begin{aligned} & |Var(\sqrt{T}P_R(T)) - \sigma^2| \\ & \leq \frac{2}{T} \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{1 - \gamma_{M,R}}{(\gamma_{M,R})^2} + \\ & \quad 2 \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \frac{(1 - \gamma_{M,R})^T}{\gamma_{M,R}} \end{aligned}$$

which concludes the proof. $\square$

Proof. Theorem 18

Firstly, by decomposing the sample variance, we derive that

$$\begin{aligned} \sqrt{\tilde{S}_R(T)} &= \sqrt{\frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B})^2} \\ &= \sqrt{\frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - E[\tilde{P}_R(T)] + E[\tilde{P}_R(T)] - \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B})^2} \\ &\leq \sqrt{2 \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - E[\tilde{P}_R(T)])^2 + \left(E[\tilde{P}_R(T)] - \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{X_t^m \in B\}} \right)^2} \\ &\leq \sqrt{2Var(\tilde{P}_R(T)) + \left(\frac{4\delta_R}{T} \right)^2} \\ &\leq \sqrt{2 \frac{Var(\sum_{t=1}^T 1_{X_t^m \in B})}{MT^2} + \left(\frac{4\delta_R}{T} \right)^2} \end{aligned}$$

Noting that there is a dependency on the true variance of $\tilde{P}_R(T)$ and the Markov chain we consider is reversible, we leverage the variance bounding technique for reversible Markov chains. It is known that for reversible, positive, geometrically ergodic Markov chains [32],

$$Var\left(\sum_{t=1}^T 1_{X_t^m \in B}\right) \leq K \cdot Var_{\pi}(1_{X_t^m \in B}) = K_R \cdot (\pi(1)(1 - \pi(1)))$$

This immediately indicates that

$$\begin{aligned} \sqrt{\tilde{S}_R(T)} &\leq \sqrt{2 \frac{Var(\sum_{t=1}^T 1_{X_t^m \in B})}{M} + \left(\frac{4\delta_R}{T} \right)^2} \\ &\leq \sqrt{2 \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2} + \left(\frac{4\delta_R}{T} \right)^2} \end{aligned}$$

which completes the first part of the statement.

Meanwhile, based on the following lemma, we derive the upper bound on K_R

Lemma 27. *Let us assume that the transition matrices of all fuzzers P_R for $R \in \mathcal{R}$ are geometrically ergodic, reversible, and positive. Then we have*

$$K_R \leq 1 + \frac{2\lambda_{R,2}}{T(1 - \lambda_{R,2})^2}$$

where $\lambda_{R,2}$ represents the spectral radius of the transition matrix P_R , denoted by the second largest eigenvalue of P_R .

Proof. The proof is straightforward by the decomposition of $Var(\sum_{t=1}^T 1_{X_t^m \in B})$. More precisely, we note that

$$\begin{aligned} & Var(T 1_{X_t^m \in B}) \\ &= \sum_{i=1}^T \sum_{j=1}^T Cov(1_{X_i^m \in B}, 1_{X_j^m \in B}) \\ &= T \cdot \gamma_0 + 2 \sum_{t=1}^{(T-1)} (T-t) \gamma_t \end{aligned}$$

where $\gamma_t = Cov(1_{X_0^m \in B}, 1_{X_t^m \in B})$.

Based on the spectral property of reversible Markov chain,

$$\gamma_t \leq \lambda_{R,2}^t \frac{1}{\pi}$$

Plugging in, we derive that

$$\begin{aligned} & Var(\sqrt{T} 1_{X_t^m \in B}) \\ &= \gamma_0 + 2 \frac{1}{T} \sum_{t=1}^{(T-1)} (T-t) \lambda_{R,2}^t \frac{1}{\pi} \\ &\leq 1 + \frac{2\lambda_{R,2}}{T(1 - \lambda_{R,2})^2} \end{aligned}$$

$\square$

We observe that the upper bound on $S_R(T)$ is monotone decreasing in the variance bound constant K_R and δ_R , as well as being monotone decreasing in $\lambda_{R,2}$. Therefore, selecting

the fuzzer with the smallest $\lambda_{R,2}$ and δ_R minimizes the upper bound, which finishes the second part of statement. $\square$

Proof. Theorem 19 The population variance reads as

$$\begin{aligned} \text{Var}(\tilde{P}_R(T)) &= \frac{1}{M} \sum_{m=1}^M (E[\tilde{P}_R(T)] - \tilde{P}_R(T))^2 \\ &= \frac{1}{M} \sum_{m=1}^M (E[\tilde{P}_R(T)] - \pi(1) + \pi(1) - \tilde{P}_R(T))^2 \\ &\leq \frac{2}{M} \sum_{m=1}^M (E[\tilde{P}_R(T)] - \pi(1))^2 + (\pi(1) - \tilde{P}_R(T))^2 \end{aligned}$$

Recall that we previously establish that

$$\begin{aligned} |E[P_R(x \in I_R^T : x \in H_R^T)] - \pi(1)| \\ \leq (1 - \pi(1)) \frac{1}{T} \frac{1}{\gamma_R} \end{aligned}$$

and

$$|\tilde{P}_R(T) - \pi(1)| \leq \sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}}$$

This immediately implies that the population variance has a finite-time upper bound on the population variance, which reads as

$$\begin{aligned} \frac{1}{M} \sum_{m=1}^M (E[\tilde{P}_R(T)] - \tilde{P}_R(T))^2 \\ \leq 2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} + 2(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}})^2 \end{aligned}$$

Meanwhile, based on the derivation of the sample variance, we obtain that

$$\begin{aligned} \text{Var}(\tilde{P}_R(T)) \\ \leq \frac{\text{Var}(\sum_{t=1}^T 1_{X_t^m \in B})}{MT^2} \\ \leq \frac{K_R \pi(1)(1 - \pi(1))}{MT^2} \end{aligned}$$

Consequently, we derive the following upper bound on the true variance

$$\begin{aligned} V_R(T) &\leq \min(2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} + \\ &2(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}})^2, \frac{K_R \pi(1)(1 - \pi(1))}{MT^2}) \end{aligned}$$

By selecting the fuzzer with parameters $\gamma_{R,2}$, δ_R , and K_R that minimizes the above quantity, we minimize the upper bound on the variance and thus select the best fuzzer. This completes the proof. $\square$

Proof. Theorem 20 By the aforementioned decomposition of $\tilde{S}_R(T)$, we obtain that

$$\begin{aligned} \tilde{S}_R(T) &= \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - \frac{1}{T} \sum_{t=1}^T 1_{X_t^m \in B})^2 \\ &\leq 2 \frac{\text{Var}(\sum_{t=1}^T 1_{X_t^m \in B})}{MT^2} + (\frac{4\delta_R}{T})^2 \end{aligned}$$

Theorem 6.1 in [4] directly suggests that

$$\begin{aligned} \text{Var}(\sum_{t=1}^T 1_{X_t^m \in B}) \\ \leq T \cdot \text{Var}(1_{X_t^m \in B}) \cdot (|C|^c + \frac{2}{\gamma_R^{MH}}) \\ \leq \pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}}). \end{aligned}$$

Consequently, we derive that

$$\tilde{S}_R(T) \leq 2 \frac{\pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}})}{MT^2} + (\frac{4\delta_R}{T})^2$$

Thus far, selecting the fuzzer that maximizes this upper bound provides the choice of best fuzzer, which completes the proof. $\square$

Proof. Theorem 21 We derive the following upper bound on the variance of the Markov chain associated with P_R through the M-H spectral gap, as introduced in [4]. Based on Theorem 6.1 in [4], we obtain the upper bound on the population variance as

$$\begin{aligned} \text{Var}(\sum_{t=1}^T 1_{X_t^m \in B}) \\ \leq T \cdot \text{Var}(1_{X_t^m \in B}) \cdot (|C|^c + \frac{2}{\gamma_R^{MH}}) \\ \leq \pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}}). \end{aligned}$$

Subsequently, the upper bound on the true variance of the estimator $\tilde{P}_R(T)$ reads as

$$\begin{aligned} \text{Var}(\tilde{P}_R(T)) \\ \leq \frac{\text{Var}(\sum_{t=1}^T 1_{X_t^m \in B})}{MT^2} \\ \leq \frac{T \cdot \text{Var}(1_{X_t^m \in B}) \cdot (|C|^c + \frac{2}{\gamma_R^{MH}})}{MT^2} \\ \leq \frac{\pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}})}{MT} \end{aligned}$$

$\square$

Proof. Theorem 22 It is worth noting that the mean square error can be decomposed as follows

$$\begin{aligned}
M_R(T) &= \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2 \\
&= \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - E[\tilde{P}_R(T)] + E[\tilde{P}_R(T)] - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2 \\
&= \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - E[\tilde{P}_R(T)])^2 + (E[\tilde{P}_R(T)] - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2 \\
&= \text{Var}(\tilde{P}_R(T)) + (E[\tilde{P}_R(T)] - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2
\end{aligned}$$

This essentially indicates that $M_R(T) = V_R(T) + B_R(T)$ where $B_R(T)$ represents the bias of the estimator $\tilde{P}_R(T)$.

Based on the aforementioned analysis, we obtain that

$$\lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T) = \pi(1)$$

and

$$\begin{aligned}
&|E[\tilde{P}_R(T)] - \pi(1)| \\
&= \left| \frac{1}{T} \sum_{t=1}^T P_R^t(1) - \pi(1) \right| \\
&\leq \frac{1}{T} \sum_{t=1}^T C_R \alpha_R^t
\end{aligned}$$

and

$$\frac{1}{T} \sum_{t=1}^T C_R \alpha_R^t \leq \min(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}})$$

Subsequently, we derive that

$$M_R(T) \leq \text{Var}(\tilde{P}_R(T)) + (\min(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}))^2$$

which depends on the true variance.

Notably, for the true variance, we establish that for reversible markov chains in Theorem 17:

$$\begin{aligned}
\text{Var}(\tilde{P}_R(T)) &\leq \min(2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} + \\
&2(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}})^2, \frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2}).
\end{aligned}$$

Subsequently, we have that for reversible Markov chains, the mean square error of $\tilde{P}_R(T)$ has the following upper bound

$$\begin{aligned}
M_R(T)^2 &\leq \text{Var}(\tilde{P}_R(T)) + (\min(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}))^2 \\
&\leq \min(2(1 - \pi(1))^2 \frac{1}{T^2} \frac{1}{\gamma_R} + 2(\sqrt{\frac{\log T}{T}} + \sqrt{\frac{4\delta_R}{T}})^2, \\
&\frac{K_R \cdot (\pi(1)(1 - \pi(1)))}{MT^2}) \\
&+ (\min(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}))^2
\end{aligned}$$

Consequently, selecting the fuzzer based on the minimal value of the upper bound leads to optimal fuzzer selection with respect to the mean square error.

This completes the proof. $\square$

Proof. Theorem 23 Likewise, we have the decomposition of $M_R(T)$ and the same upper bound on the second bias term $B_R(T)$.

$$\begin{aligned}
M_R(T) &= \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2 \\
&= \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - E[\tilde{P}_R(T)] + E[\tilde{P}_R(T)] - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2 \\
&= \frac{1}{M} \sum_{m=1}^M (\tilde{P}_R(T) - E[\tilde{P}_R(T)])^2 + (E[\tilde{P}_R(T)] - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2 \\
&= \text{Var}(\tilde{P}_R(T)) + (E[\tilde{P}_R(T)] - \lim_{T \rightarrow \infty} P_R(x \in I_R^T : x \in H_R^T))^2
\end{aligned}$$

Subsequently, we derive that

$$M_R(T) \leq \text{Var}(\tilde{P}_R(T)) + (\min(C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}}))^2$$

which depends on the true variance.

Notably, for the true variance, we establish that for non-reversible Markov chains in Theorem 14:

$$\text{Var}(\tilde{P}_R(T)) \leq \frac{\pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}})}{MT}.$$

Consequently, putting everything together, we derive the following upper bound on the estimator $M_R(T)$

$$\begin{aligned}
M_R(T) &= \text{Var}(\tilde{P}_R(T)) + \\
&\left(\min \left\{ C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}} \right\} \right)^2 \\
&\leq \frac{\pi(1)(1 - \pi(1))(|C|^c + \frac{2}{\gamma_R^{MH}})}{MT} + \\
&\left(\min \left\{ C_R \frac{1}{T} \frac{1}{1 - \alpha_R}, \sqrt{\frac{4\delta_R}{T}} \right\} \right)^2.
\end{aligned}$$

□ Notably, assuming $\mathbb{P}(R(x, w) \in B) \geq \delta$ allows us to define $\beta(x) := \frac{1}{\mathbb{P}(R(x, w) \in B)}$, and this implies that

$$R_v \geq \frac{\delta}{\delta_1}.$$

11. Section 6.1.1

Proof. Theorem 6

Define the (diagonal) second-moment term

$$d_{ii}(x) := \mathbb{E}[\hat{h}_{ii}(x)^2] - \left(\mathbb{E}[\hat{h}_{ii}(x)]\right)^2. \quad (2)$$

Define the normalized weight

$$\tau(x) := \frac{\pi(x) \bar{w}(x)}{\int \pi(u) \beta(u) \bar{w}(u) du}. \quad (3)$$

Let the branching density be

$$\beta(x) := \frac{1}{\bar{w}(x)} \quad (\text{or } 0 \text{ if } w = \infty), \quad w(x) = \min_j \hat{p}_j \cdot \hat{P}_R(x \in B). \quad (4)$$

With the above notation,

$$R_v := \frac{\det(\mathcal{D}(\tilde{w})(T))}{\det(\mathcal{D}(\mathbf{1})(T))} = \frac{\int \tau^{-1}(x) d_{ii}(x) \pi(x) dx}{\int d_{ii}(x) \pi(x) dx}. \quad (5)$$

When $\pi(x) = 1/|D|$, (3) becomes

$$\tau(x) = \frac{\pi(x) \bar{w}(x)}{\int \pi(u) \frac{1}{w(u)} du} = \frac{w(x)}{|D|} \int \frac{1}{w(u)} du. \quad (6)$$

Moreover, if $d_{ii}(x) = \text{Var}_T(\hat{h}_{ii}(x))$ is independent of x , i.e.,

$$d_{ii}(x) = \text{Var}_T(\hat{h}_{ii}(x)) = \text{Var}\left(\int g p(x)\right), \quad \text{independent of } x, \text{ where} \quad (7)$$

then (5) reduces to

$$\begin{aligned} R_v &= \int \tau^{-1}(x) \pi^{-1}(x) dx = \frac{1}{|D|^2} \int \tau^{-1}(x) dx \\ &= \frac{\int \beta(x) dx}{|D|^2} \int \frac{1}{\beta(x)} dx. \end{aligned} \quad (8)$$

Consequently, we derive that

$$\begin{aligned} R_v &= \frac{\int \beta(x) dx}{|D|^2} \int \frac{1}{\beta(x)} dx \\ &= \frac{\int \frac{1}{w(x)} dx}{|D|^2} \int w(x) dx. \end{aligned}$$

where $w(x) = \frac{1}{\beta(x)}$.

Let us consider that

$$w(x) = \mathbb{P}(R(x, w) \in B).$$

Assume $\delta \leq \mathbb{P}(R(x, w) \in B) \leq \delta_1, \forall x$, (hence $\delta \leq \delta_1$).

$$\int \left(\frac{1}{\mathbb{P}(R(x, w) \in B)}\right)^a dx \geq \frac{1}{|\delta_1|^a}.$$

Proof. Theorem 7 By the definition of R_v , we have that

$$\begin{aligned} R_v &= \frac{\int \beta(x) dx \int \frac{1}{\beta(x)} dx}{|D|^2} \\ &\geq \frac{\int \beta(x) dx \int \omega(x) dx}{|D|^2} \\ &\geq \frac{\int \beta(x) dx \int \mathbf{1}_{\{R(x, w) \in B\}} dx}{|D|^2} \\ &= \frac{\int \beta(x) dx \mathbb{P}(R(x, w) \in B)}{|D|} \\ &= \left[a \mathbb{P}(R(x, w) \notin B) + \mathbb{P}(R(x, w) \in B) \right] \mathbb{P}_x((x, w) \in B) \\ &= a \mathbb{P}(R(x, w) \notin B) \mathbb{P}_x((x, w) \in B) + \mathbb{P}(R(x, w) \in B)^2. \end{aligned}$$

Choose a and x such that

$$\begin{aligned} &a \mathbb{P}(R(x, w) \notin B) \mathbb{P}_x((x, w) \in B) + \mathbb{P}(R(x, w) \in B)^2 \\ &= \max_a \min_x a \mathbb{P}(R(x, w) \notin B) \mathbb{P}_x((x, w) \in B) + \mathbb{P}(R(x, w) \in B)^2. \end{aligned}$$

This gives us that

$$R_v \geq V,$$

$$\begin{aligned} V &= \max_a \min_x a \mathbb{P}(R(x, w) \notin B) \mathbb{P}_x((x, w) \in B) \\ &\quad + \mathbb{P}(R(x, w) \in B)^2. \end{aligned}$$

This demonstrates the efficiency effectiveness of the splitting algorithm compared to the optimal splitting measure. □

12. Section 6.2

Setup.. Recall $Y_{k,t} = \mathbf{1}\{X_{k,t} \in B\}$ for post-split replications $k \in \{1, 2\}$ and $Y_{s,t} = \mathbf{1}\{X_{s,t} \in B\}$ for the shared pre-split trajectory. Under one split at time τ , we have $N_t = 1$ for $t \leq \tau$ and $N_t = 2$ for $t > \tau$, so the splitting estimator can be written as

$$\hat{P}_R(T) = \frac{1}{T} \left(\sum_{t=1}^{\tau} Y_{s,t} + \sum_{t=\tau+1}^T \frac{1}{2} \sum_{k=1}^2 Y_{k,t} \right). \quad (9)$$

At the split time τ , the two post-split trajectories start from the same state (the pre-split state at time τ), and all post-split randomness is generated independently of the pre-split randomness conditional on $X_{s,\tau}$. We view τ as a regeneration point and treat the pre- and post-split contributions as uncorrelated at the level of the variance decomposition.

Step 1: Variance decomposition.. Since $Y_{s,\tau}$ is fixed given the split event, it does not contribute to the variance. Using (9) and the regeneration-cycle viewpoint at τ , we decompose

$$\begin{aligned} Var(\hat{P}_R(T)) &= Var\left(\frac{1}{T} \sum_{t=\tau+1}^T \frac{1}{2} \sum_{k=1}^2 Y_{k,t} + \frac{1}{T} \sum_{t=1}^{\tau-1} Y_{s,t}\right) \\ &= \frac{1}{4T^2} Var\left(\sum_{t=\tau+1}^T (Y_{1,t} + Y_{2,t})\right) + \frac{1}{T^2} Var\left(\sum_{t=1}^{\tau-1} Y_{s,t}\right). \end{aligned} \quad (10)$$

Expanding the first term yields

$$\begin{aligned} &Var\left(\sum_{t=\tau+1}^T (Y_{1,t} + Y_{2,t})\right) \\ &= Var\left(\sum_{t=\tau+1}^T Y_{1,t}\right) + Var\left(\sum_{t=\tau+1}^T Y_{2,t}\right) + \\ &2Cov\left(\sum_{t=\tau+1}^T Y_{1,t}, \sum_{t=\tau+1}^T Y_{2,t}\right). \end{aligned} \quad (11)$$

By symmetry of the two post-split replications, $Var(\sum Y_{1,t}) = Var(\sum Y_{2,t})$.

Step 2: Controlling the post-split covariance via the multiplicative correlation bound.. Using the correlation control condition

$P(X_{1,t} \in B, X_{2,t} \in B) \leq (1-\rho)P(X_{1,t} \in B)P(X_{2,t} \in B)$, we obtain for each $t > \tau$,

$$\begin{aligned} Cov(Y_{1,t}, Y_{2,t}) &= P(X_{1,t} \in B, X_{2,t} \in B) - P(X_{1,t} \in B)P(X_{2,t} \in B) \\ &\leq -\rho \cdot P(X_{1,t} \in B)P(X_{2,t} \in B). \end{aligned} \quad (12)$$

Summing over $t = \tau + 1, \dots, T$ gives

$$\begin{aligned} &Cov\left(\sum_{t=\tau+1}^T Y_{1,t}, \sum_{t=\tau+1}^T Y_{2,t}\right) \\ &= \sum_{t=\tau+1}^T Cov(Y_{1,t}, Y_{2,t}) \\ &\leq -\rho \sum_{t=\tau+1}^T P(X_{1,t} \in B)P(X_{2,t} \in B). \end{aligned} \quad (13)$$

Plugging (13) into (10)–(11) yields

$$\begin{aligned} &Var(\hat{P}_R(T)) \\ &= \frac{1}{2T^2} Var\left(\sum_{t=\tau+1}^T Y_{1,t}\right) + \frac{1}{T^2} Var\left(\sum_{t=1}^{\tau-1} Y_{s,t}\right) - \\ &\frac{\rho}{2T^2} \sum_{t=\tau+1}^T P(X_{1,t} \in B)P(X_{2,t} \in B), \end{aligned} \quad (14)$$

which is the claimed variance comparison form in Section 6.3.2.

Step 3: Asymptotic order and the G_T term.. Let $f(x) = \mathbf{1}\{x \in B\}$ and define the asymptotic variance

$$\sigma_f^2 = \lim_{T \rightarrow \infty} Var\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T f(X_t)\right).$$

For a geometrically ergodic chain, $Var(\sum_{t=\tau+1}^T Y_{1,t}) = (T-\tau)\sigma_f^2 + o(T)$. If additionally τ exceeds the mixing time, then $P(X_{1,t} \in B) \approx P(X_{2,t} \in B) \approx \pi(1)$ for $t > \tau$, and the negative term in (14) scales as

$$\begin{aligned} &\frac{\rho}{2T^2} \sum_{t=\tau+1}^T P(X_{1,t} \in B)P(X_{2,t} \in B) \\ &\approx \frac{\rho}{2T^2} (T-\tau)\pi(1)^2 = O\left(\frac{\rho}{T}\right). \end{aligned}$$

Collecting the dominant terms yields the stated G_T scaling:

$$Var(\hat{P}_R(T)) - Var(\tilde{P}_R(T)) \approx G_T = \frac{1}{2T}(\sigma_f^2 - \rho).$$

Step 4: Closed-form expressions in the stationary regime.. When the chain is stationary with invariant distribution π , the standard variance formula gives

$$\sigma_f^2 = Var_\pi(f) + 2 \sum_{k=1}^{\infty} Cov_\pi(f(X_0), f(X_k)).$$

For a two-state (bug / non-bug) abstraction with spectral gap $\gamma_{M,R}$ and $f = \mathbf{1}\{X \in B\}$, one obtains

$$\begin{aligned} Var_\pi(f) &= \pi(1)\pi(0), \\ Cov_\pi(f(X_0), f(X_k)) &= \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} (1 - \gamma_{M,R})^k, \end{aligned}$$

and summing the geometric series yields

$$\sigma_f^2 = \pi(1)\pi(0) + \frac{\pi(1)}{2 \max\{\pi(0), \pi(1)\}} \cdot \frac{1 - \gamma_{M,R}}{\gamma_{M,R}}.$$

Next, under the AR(1) masking/mutation model, the correlation constant ρ can be interpreted as the fractional decrease in joint bug probability induced by the coupled construction. In the simplified discrete model in which (i) the mutation value is uniform over a finite input domain of size $|D|$, and (ii) the event $\{X_t \in B\}$ is triggered by a single location/value pattern, the resulting expression reduces to

$$\rho = \frac{1}{|D| \cdot |\pi(1)|}.$$

Step 5: Non-stationary extension (controlled deviation).. When the chain is non-stationary, assume the deviation between the stationary distribution π and the marginal distribution at time t (after a burn-in) is bounded by δ in total variation, and the mapping from this deviation to ρ is L -Lipschitz. Then

$$|\rho - \frac{1}{|D| \cdot |\pi(1)|}| \leq L\delta,$$

which yields the stated bounds on ρ used in Section 6.3.3.

13. Variance Comparison between Splitting and MC

Proof of Theorem 2. Non-splitting (vanilla MC). Let $\tilde{\mu}_T = \frac{1}{T} \sum_{t=1}^T f(X_t)$ with X_t a stationary Markov chain with kernel P . Then

$$\text{Var}(\tilde{\mu}_T) = \frac{\sigma_{\text{asym}}^2(f)}{T} + o(T^{-1}), \quad \text{CPU}_{\text{MC}}(T) = T,$$

and hence

$$\mathcal{E}_{\text{MC}}(T) = \text{Var}(\tilde{\mu}_T) \cdot T = \sigma_{\text{asym}}^2(f) + o(1).$$

Splitting estimator (two descendants; re-splitting allowed).

Let $X_t^{(1)}, \dots, X_t^{(k_t)}$ be the active paths at time t , and

$$\bar{Z}_t := \frac{1}{k_t} \sum_{i=1}^{k_t} f(X_t^{(i)}), \quad \hat{\mu}_T := \frac{1}{T} \sum_{t=1}^T \bar{Z}_t, \quad \text{CPU}_{\text{split}}(T) = \sum_{t=1}^T k_t.$$

Define the (finite-horizon) local autocovariance weight

$$c_t^{(T)} := \text{Var}(g(X_t)) + 2 \sum_{\ell=1}^{T-t} |\text{Cov}(g(X_t), g(X_{t+\ell}))|,$$

$$g := f - \pi(f).$$

Under Assumption A1, we have

$$\text{Var}(\hat{\mu}_T) \leq \frac{1}{T^2} \sum_{t=1}^T \frac{c_t^{(T)}}{k_t},$$

$$\mathcal{E}_{\text{split}}(T) \leq \frac{1}{T^2} \left(\sum_{t=1}^T k_t \right) \left(\sum_{t=1}^T \frac{c_t^{(T)}}{k_t} \right).$$

Two-level allocation comparison ($k_t \in \{1, 2\}$). Based on our splitting algorithm,

$$k_t = \begin{cases} 2, & X_t \in B, \\ 1, & X_t \notin B. \end{cases}$$

Let $S := \{t \in [T] : X_t \in B\}$, and define

$$C_{\text{in}} := \sum_{t \in S} c_t^{(T)}, \quad C_{\text{out}} := \sum_{t \notin S} c_t^{(T)}.$$

Then we obtain

$$\mathcal{E}_{\text{split}}(T) \leq \frac{T + |S|}{T^2} \left(C_{\text{out}} + \frac{1}{2} C_{\text{in}} \right),$$

$$\mathcal{E}_{\text{MC}}(T) \text{ (via the same bound with } k_t \equiv 1) \leq \frac{1}{T} (C_{\text{out}} + C_{\text{in}}).$$

Therefore,

$$\frac{\mathcal{E}_{\text{split}}(T)}{\mathcal{E}_{\text{MC}}(T)} \leq \frac{T + |S|}{T} \cdot \frac{C_{\text{out}} + \frac{1}{2} C_{\text{in}}}{C_{\text{out}} + C_{\text{in}}}.$$

This implies that splitting improves variance-per-CPU whenever C_{in} (the autocovariance mass contributed by times in B) dominates C_{out} . In the extreme clustered regime $C_{\text{in}} \gg C_{\text{out}}$, the ratio approaches

$$\frac{\mathcal{E}_{\text{split}}(T)}{\mathcal{E}_{\text{MC}}(T)} \approx \frac{T + |S|}{T} \cdot \frac{1}{2}.$$

Under stationarity, $|S|/T \approx p = \pi(B)$, recovering the heuristic $(1+p)/2$. $\square$

14. Bug Detection Rate of Splitting

Proof of Theorem 3. Let R_{MC} and R_{split} denote the bug-detection rates of MC and Splitting, respectively. To avoid notational collision, it is convenient to view $\{Y_{s,t}\}$ as the bug indicators produced by Splitting at (outer) time t , where $s \in \{1, \dots, k_t\}$, and to identify the MC indicator at time t with $Y_{1,t}$. Then

$$R_{\text{MC}} = \frac{1}{T} \sum_{t=1}^T Y_{1,t} = \frac{|S_{\text{MC}}|}{T}.$$

Under Splitting, since $k_t \in \{1, 2\}$ and $k_t = 2$ exactly on the splitting times, define

$$A := \{t \in [T] : k_t = 2\}.$$

Then the Splitting rate can be written as

$$R_{\text{split}} = \frac{1}{T} \sum_{t=1}^T \frac{1}{k_t} \sum_{s=1}^{k_t} Y_{s,t} = \frac{1}{T} \sum_{t \notin A} Y_{1,t} + \frac{1}{2T} \sum_{t \in A} (Y_{1,t} + Y_{2,t}).$$

Therefore, the increment of Splitting over MC is

$$\begin{aligned} R_{\text{split}} - R_{\text{MC}} &= \left[\frac{1}{T} \sum_{t \notin A} Y_{1,t} + \frac{1}{2T} \sum_{t \in A} (Y_{1,t} + Y_{2,t}) \right] - \frac{1}{T} \sum_{t=1}^T Y_{1,t} \\ &= \frac{1}{2T} \sum_{t \in A} (Y_{2,t} - Y_{1,t}). \end{aligned}$$

This is an exact identity.

Using your relation

$$|S_{\text{splitting}}| - |S_{\text{MC}}| = \sum_{t=1}^T \sum_{s=2}^{k_t} Y_{s,t} = \sum_{t \in A} Y_{2,t},$$

we can equivalently rewrite the increment as

$$R_{\text{split}} - R_{\text{MC}} = \frac{1}{2T} \left(|S_{\text{splitting}}| - |S_{\text{MC}}| - \sum_{t \in A} Y_{1,t} \right).$$

As a simple consequence (since $0 \leq Y_{1,t} \leq 1$),

$$\frac{1}{2T} \left(|S_{\text{splitting}}| - |S_{\text{MC}}| - |A| \right) \leq R_{\text{split}} - R_{\text{MC}} \leq \frac{1}{2T} \left(|S_{\text{splitting}}| - |S_{\text{MC}}| \right),$$

and $R_{\text{split}} \geq R_{\text{MC}}$ holds whenever

$$\sum_{t \in A} Y_{2,t} \geq \sum_{t \in A} Y_{1,t}.$$

$\square$